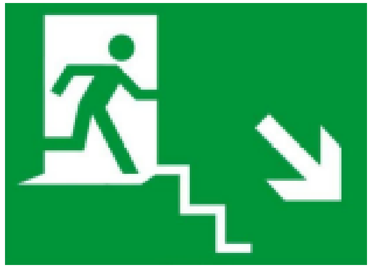
# **FIRE** *the weekly fire alarm test is 10am on Wednesdays*

- Follow green ‘running man’ signs to the fire escape routes signposted ‘Fire Exit’ with an arrow pointing to the nearest exit
- Close doors behind you
- Let us know if you need assistance to leave the building
- Assemble on the grassy area at the front of the Nucleus until told it is safe to re-enter the building




**DON’T**

*Welcome to*

# Dealing With Data

SBS PhD Induction - Autumn 2025

***Alli Spring, PhD.***
***Daniel Thédié, PhD.***
***Juliana Rodríguez Cubillos***

**bio_rdm@ed.ac.uk**

# Course overview

- [FAIR principles for managing data]
- [Managing FINDABLE data]
- [Managing ACCESSIBLE data]
- [Managing INTEROPERABLE data]
- [Managing REUSABLE data]
- [Data-related resources at the University (UoE Digital Research Services team)]

# FAIR principles for managing data

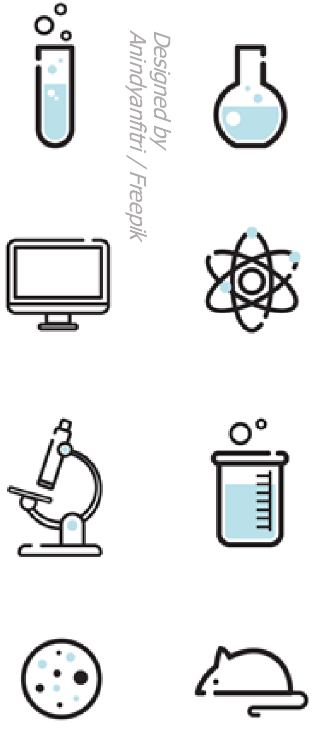
FAIR
framework

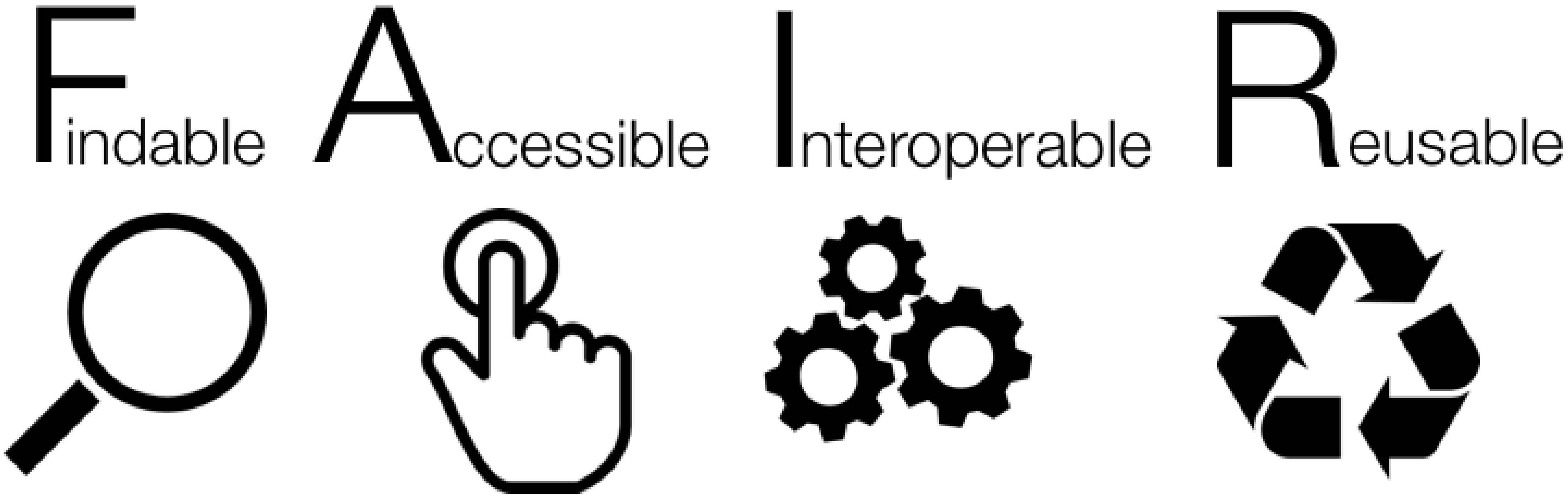
# What is data?

Data can be defined as facts, figures, or information that is collected, used and stored for reference or analysis

*Biological data is diverse!*

- **Information about physical samples**:
  - animal/ plant (live, cell cultures, frozen tissues, etc)
  - patient information
  - bacterial / yeast stocks
- **Quantitative / numerical data**: Spreadsheets (tables), csv files
- **Images** (pictures, microscopy images, 3D scans)
- **Code** (Software programs, analysis codes, algorithms)
- **DNA, RNA, protein sequences** (plasmids, NGS sequencing files, etc)
- **Protocols** (recipes, laboratory and measurement protocols)






Wilkinson *et al.* The FAIR Guiding Principles for scientific data management and stewardship. *Sci Data* **3**, 160018 (2016). https://doi.org/10.1038/sdata.2016.18

FAIR framework

# Common Issues in Data Sharing

Only the averaged data is available.

No numerical data available.

Data tables as PDF files in the supporting information.

Vendor-specific file formats.

Links to non-existing group websites/databases.

Data / Code "on request".

# Managing FINDABLE data

What makes data findable, and why does it matter?

- ✓ [Metadata]
- ✓ [File Naming]
- ✓ [Data Structure]

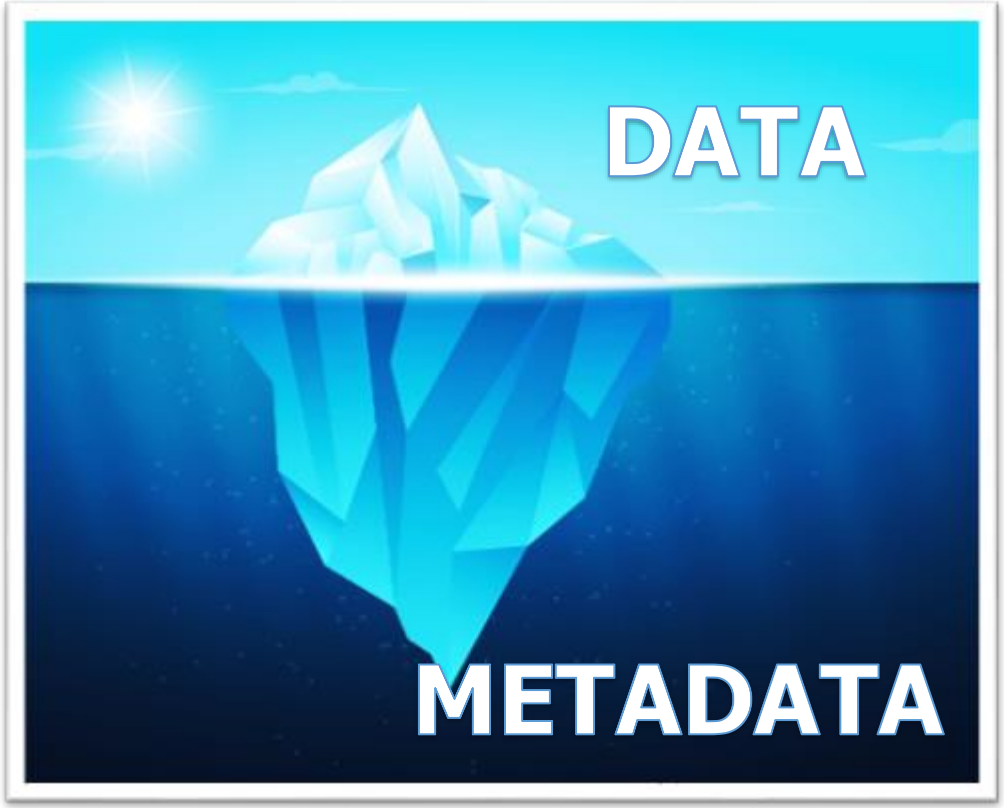
# Metadata

# What is metadata?

Metadata is the data about data!

Description/information of the data

The data has little or no meaning without metadata




"Iceberg image designed by pikisuperstar / Freepik"

✓ It should be continuously added to your research data (not just at the beginning or end of a project!)

✓ Metadata plays a significant role in making your data **FAIR.**

What information would you need to re-use this data?

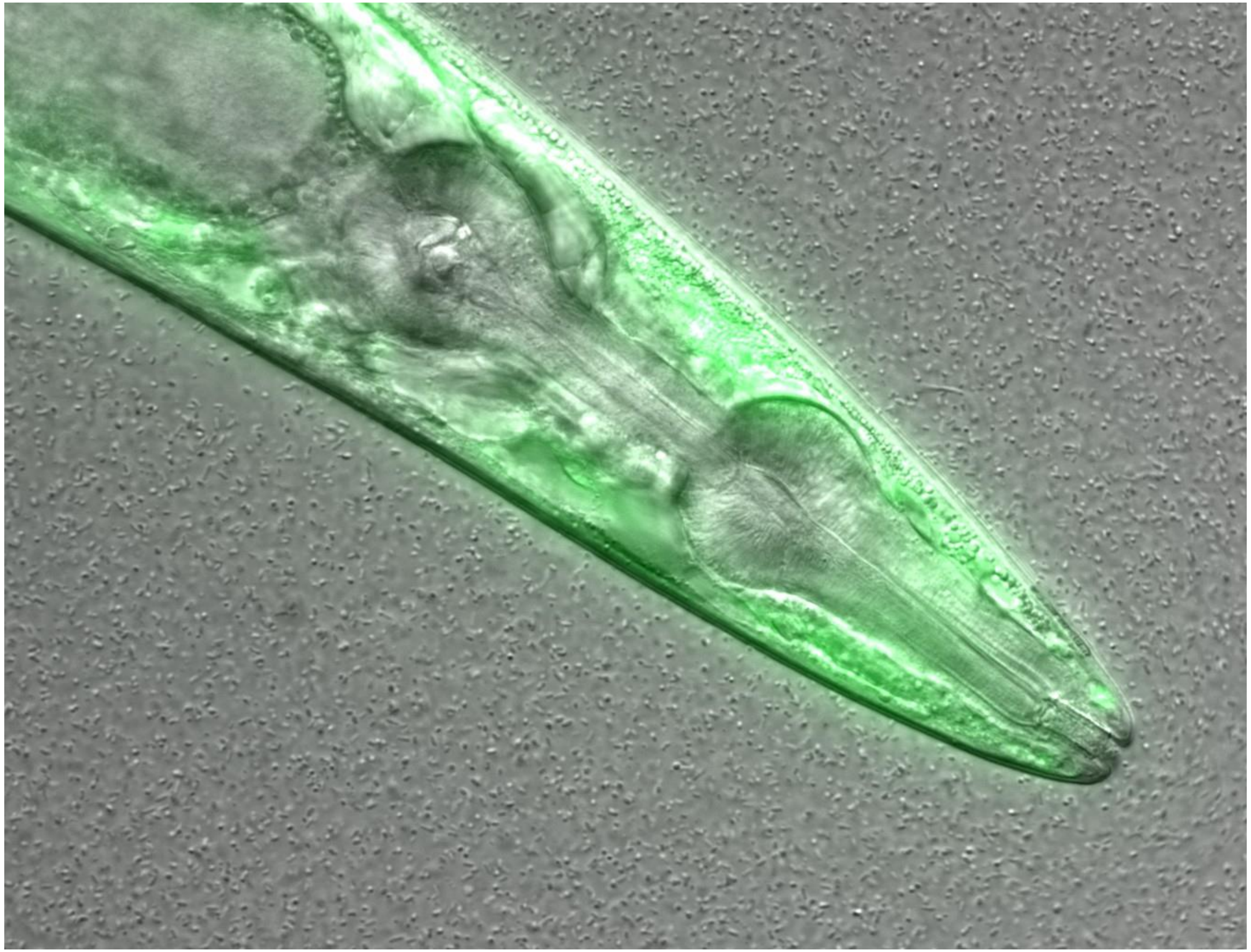


*Figure credits: María Eugenia Goya*

# File Naming

# File Naming

## Naming + Organisation Goals

Find files quickly

Ability to tell the file content without opening it

## Standardised Naming Conventions are…

- ✓ Consistent.
- ✓ Encoding of experimental factors.
- ✓ Meaningful to you and your collaborators.
- ✓ Give you a sense of the content.
- ✓ Easily identify if something is missing.

# File Naming

File names include:

- Date → *ISO format: YYYY-MM-DD or YYYYMMDD (e.g. 20241106)**
- Version Number → *Use leading zeroes (e.g. v005 instead of v5)*
- Prefix for Numerical Value → *e.g. S003, TR005*
- Relevant Biological or Experimental Info → *Sample, Site, Patient ID, Drug treatments, Doses, Names, Environmental conditions, Genotypes, Markers, Technique, Data operation (normalized, cleaned, detrended, clustered)*

* Add date at the end of the file *UNLESS* you organize files chronologically

# File Naming

Findable file names also include:

- File Format Extension → *e.g. .docx, .xlsx, .mov, .tif*
- PROJECT_STRUCTURE (README) file → *naming convention, directory structure and abbreviations*

# File Naming

An excellent example:

**LD_phyA_off_t04_2020-08-12_norm.xlsx**

We know that the file is…

- from an experiment long day (LD).
- for genotype phyA.
- with media without sucrose (off).
- at timepoint 4 (t04).
- normalised data (norm).

# File Naming

Best to avoid the following:

- Spaces → *use _ or - instead*
- Commas + Special Characters → *e.g. ~ ! @ # $ % < > ?[ ] { } '*
- Language-specific characters → *e.g. ó ę ż ï ň ũ*
- Long names → *fewer than 30 characters*
- Deep Paths with Long Names → *i.e. deeply nested folders with long names*
- Parent Element Info →

    *ELN_MI_20200101.tiff in Electron_Microscopy folder*

# Data Structure

# Project organization

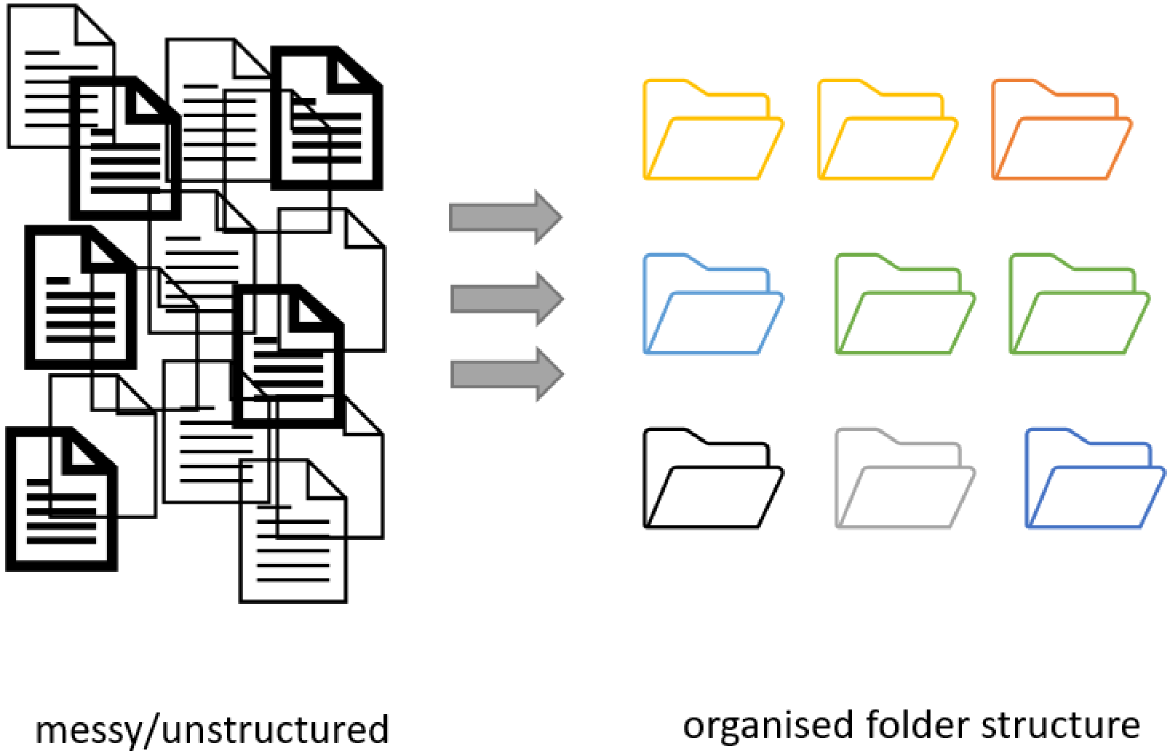



*Figure credits: Andrés Romanowski*

## *Folders vs Files*

- Folders permit grouping relevant data together
- Folders help to keep files names short

# Exercise #1

- **Resource**: Trainee's Personal Data Files or Provided Data Files
  - bit.ly/4ny6ULC

- **Task A**: Edit file names and folder structure according to best practices
- **Task B**: Edit the ReadMe file with the following info:
  - Folder description
  - Author(s) info
  - Description of folder contents
  - Naming conventions used

- **Output**: Basic folder stub

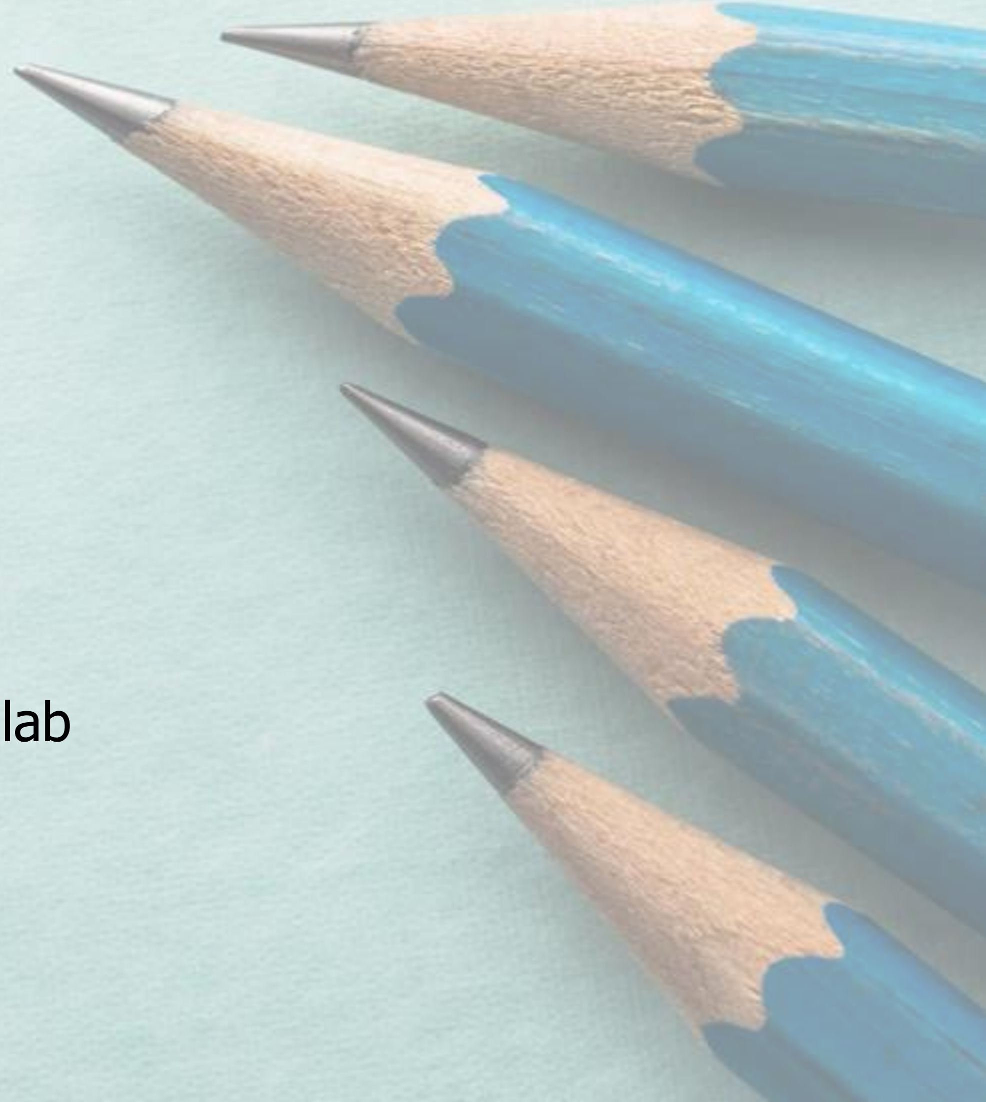
**Meaningful_name_date.extension**

# Enduring Understanding

Data findability is impacted by file naming, metadata, and folder structure. (Meta)data that cannot be found cannot be reused.

# Practical Applications

- Understand your own work in 2 years time!
- Make it easier to pass on your work when you leave the lab
- Make your data sharing-ready



# Managing **ACCESSIBLE** data

To what extent is data accessibility contingent on
Open practice and the FAIR principles?

- ✓ Repositories
- ✓ Electronic Notebooks
- ✓ Jupyter Notebooks

# Repositories

# Repositories

- Deposit data to an external, reputable public repository.
- Repositories provide **persistent identifiers** (PIDs), catalogue options, advanced metadata searching, and download statistics.
- Uploading data to a repository **requires rich metadata**

# Repositories

Some repositories can also host private data or provide embargo periods, meaning access to all data can be delayed.

There are general “data agnostic” repositories, for example: Dryad, Zenodo, FigShare and THE UNIVERSITY’s own DataShare .

Or domain specific, for example: UniProt – protein data, GenBank – sequence data, MetaboLights – metabolomics data.

Accessible

# Electronic Lab Notebooks

Accessible

# Good Lab Records

| | |
|---|---|
| **Laboratory Notebooks** | Contain all relevant details (what, who, when why and how) |
| | Which project(s) is the record part of |
| | Information on lot/batch numbers |
| | What happened and what did not happen (data, including images) |
| | How you processed and analysed the data |
| | Your interpretation (and the interpretations of others if important) and next steps in the project based on these results |
| | Should be well organised for ease of navigation (indexed, labelled, catalogued) |
| | Accurate and complete: include all original data and important study details (metadata). |

REMEMBER!

Both protocols and laboratory records need to be detailed, accurate and complete.

# Good Lab Records

| | |
|---|---|
| **Protocols** | Give credit to the person who created the protocol, if it was not you. |
| | Complete and detailed instructions describing why and how to do an experiment |
| | What special materials and instruments are being used and where they were obtained |
| | Health and safety advice and how to dispose of waste |
| | Allow repetition of your procedures and studies by yourself and others |


Figure credits @sketchscience

# ELNs

*Let's make ourselves a cup of tea.*

Here's a peer-reviewed protocol for making tea

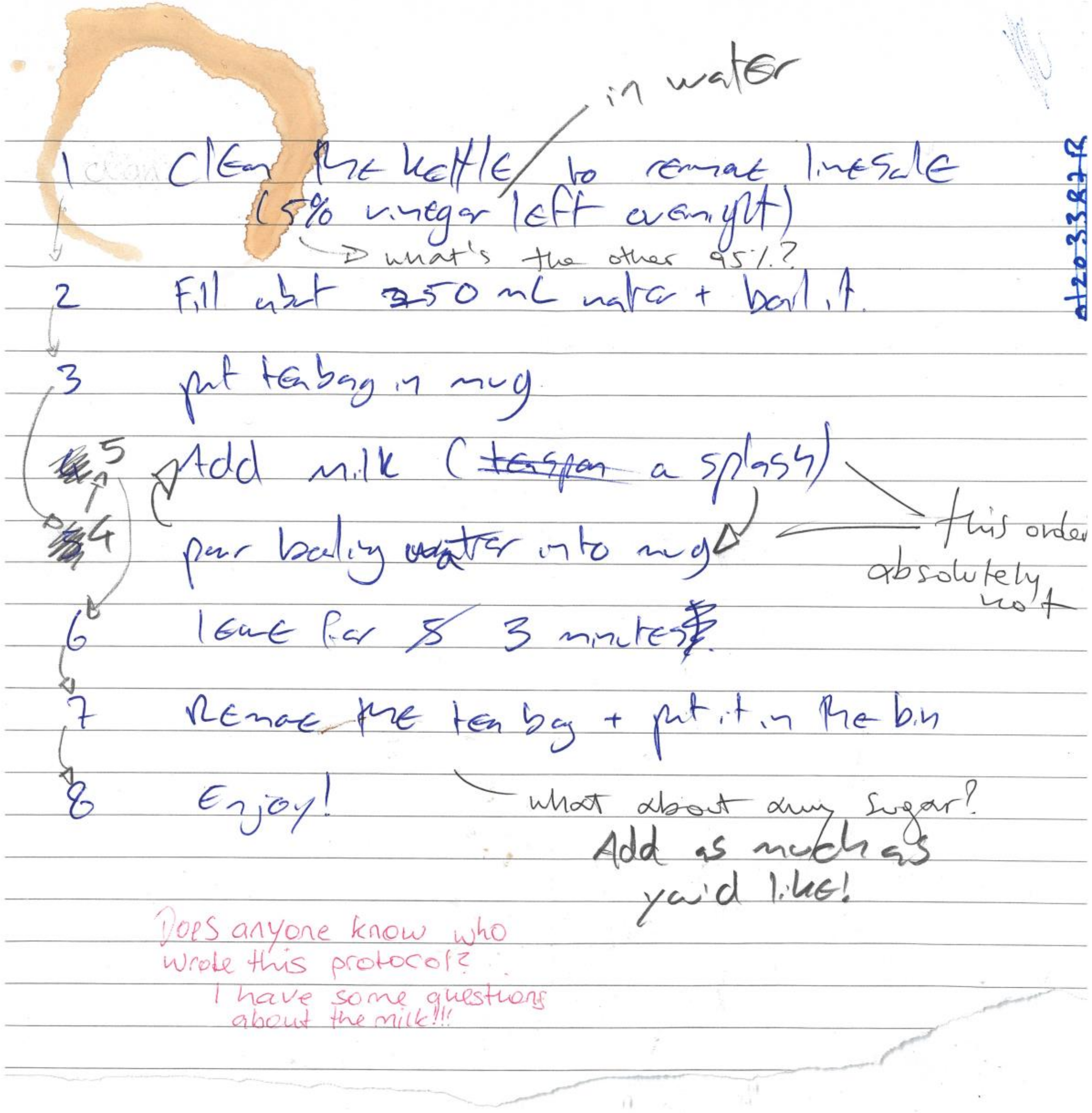
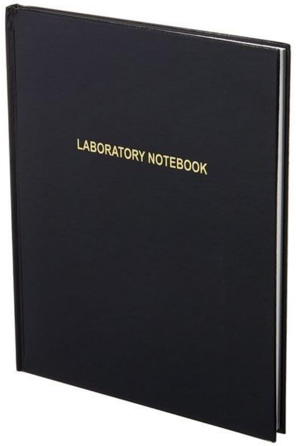

1 ~~clean~~ Clean the kettle / in water to remove limescale (5% vinegar left overnight)
→ what's the other 95%?
2 Fill abt ~~2~~50 mL water + boil it.
3 put teabag in mug
~~4~~ 5 Add milk (~~teaspoon~~ a splash)
~~5~~ 4 pour boiling ~~water~~ into mug
this order absolutely not
6 leave for ~~5~~ 3 minutes
7 Remove the tea bag + put it in the bin
8 Enjoy!
what about any sugar?
Add as much as you'd like!
Does anyone know who wrote this protocol?
I have some questions about the milk!!!

*Figure credits: Dr Ines Boehm and Ben Thomas*

Accessible

# Paper



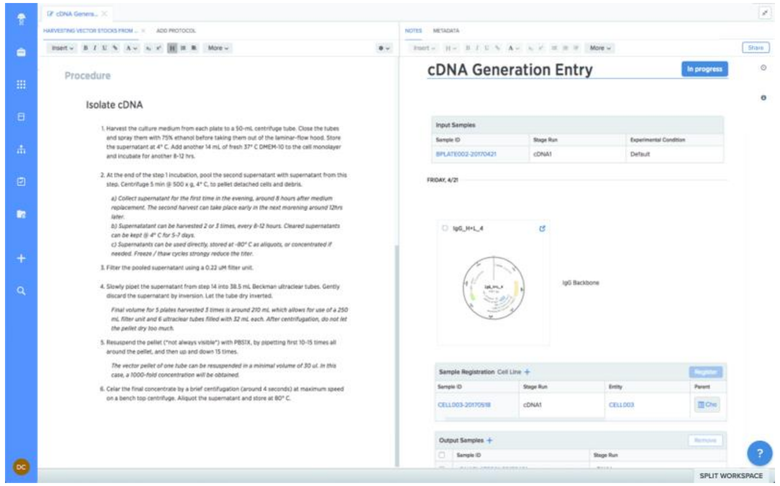

- It does not require the internet.
- Directly draw on your records.

Pros

- Can be lost or damaged
- Handwriting is not always intelligible
- Hard to edit and share
- Hard to reuse information
- Not findable/ accessible

Cons

# ELN




- Your data is organised and searchable.
- Allows off-site archiving.
- You can have templates (e.g.: protocols/assay / entries ) - Easy to copy data entries
- Can be shared instantly; global access within a group.
- Doesn’t take up physical space (no record rooms/folders)
- Regular backups.

Pros

- Requires internet access
- Better if you have a laptop or tablet

Cons

# How do I choose an ELN?

Does my group already use one?

**Yes**

Use that one!

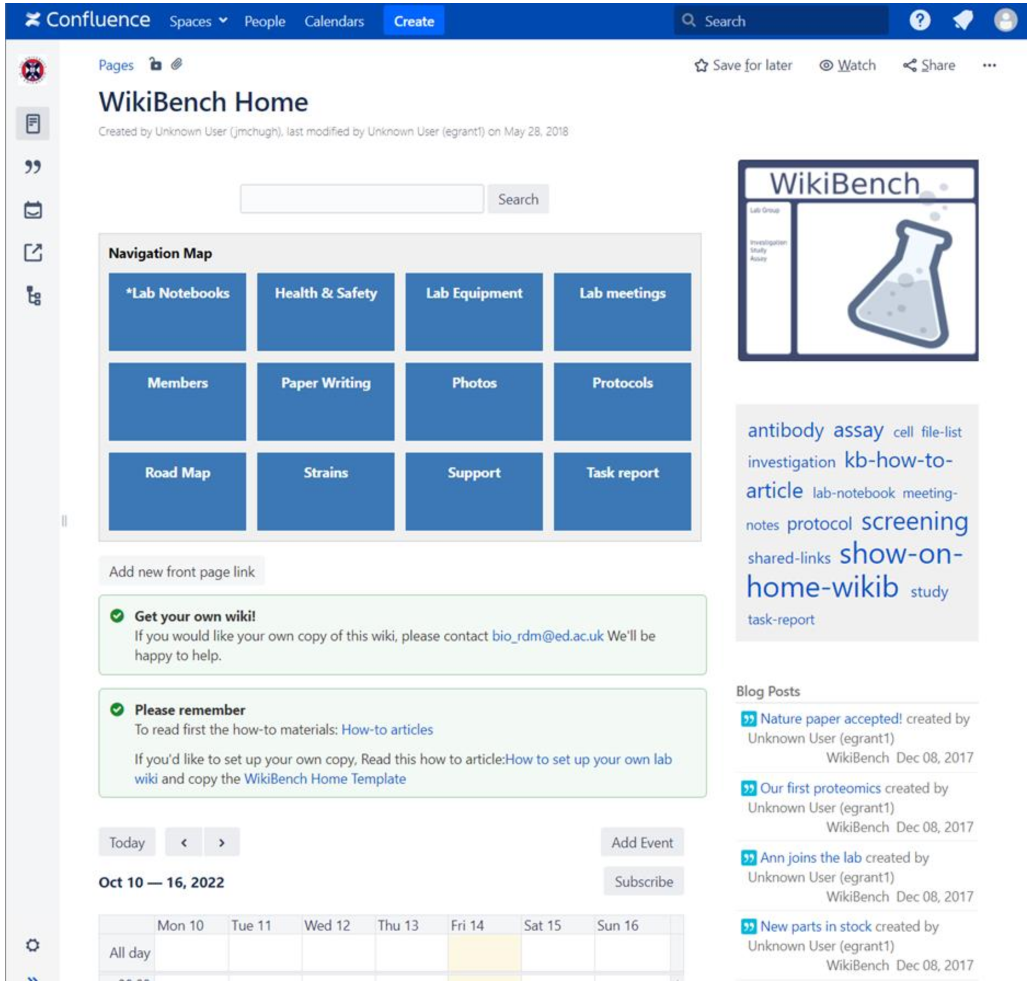
**No**

Things to consider:

- Cost
- Traceable? (data cannot be deleted and is time-stamped)
- Where are the servers located?
- Can the data be downloaded?
- What happens to the data when someone leaves the group?

# Which ELNs are used in SBS?

## WikiBench (UoE Wiki)




Benchling

RSpace

labarchives
Better Science

jupyter

For code

protocols.io

For keeping/sharing protocols

Accessible

# Example of good record keeping

Aims of the experiment

Samples are described

Protocols are written down and referenced

Result figures and description

What worked and what did not; potential solutions

Final conclusions

Issues accessing the link? Please let us know ☺

Benchling

https://benchling.com/s/etr-0FdV1H0rpWeHk4H72NOg/edit

# ELNs Resources from BioRDM

https://www.wiki.ed.ac.uk/x/f0SkGw

Hands-on tutorial on Benchling:

https://www.wiki.ed.ac.uk/spaces/RDMS/pages/463750251/Benchling+quick+tutorial

Benchling

bio_rdm@ed.ac.uk

# Demo

# Exercise #2

**Resource**: Wet lab protocol or Piece of Code

- bit.ly/42lszhq

**Task Option A**: Create a protocol on an ELN

- benchling.com

**Task Option B**: Transform a piece of code into a Jupyter notebook and add comments to it

- mango.bio.ed.ac.uk/jupyter
- Password: FAIR2025

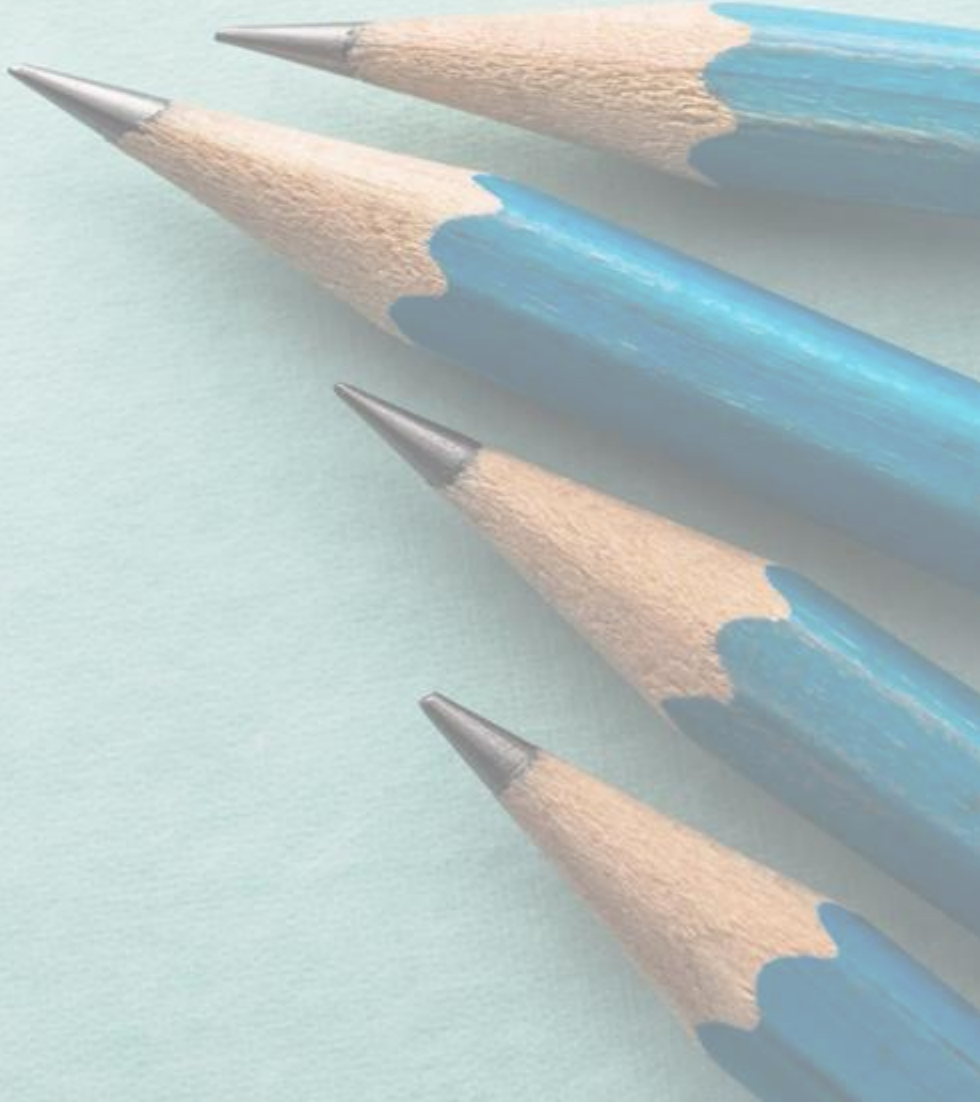
**Outputs**: Registered ELN account and protocol stub or commented Jupyter Notebook

# Enduring Understanding

Data accessibility is facilitated via ELNs, file types, and repositories. Embedded levels of openness (i.e. format, file type, publishing) make data accessible, not simply openness of FAIR practice alone.

# Practical Applications

- Facilitate work within your team
- Keep reliable records of your work
- Find information quickly



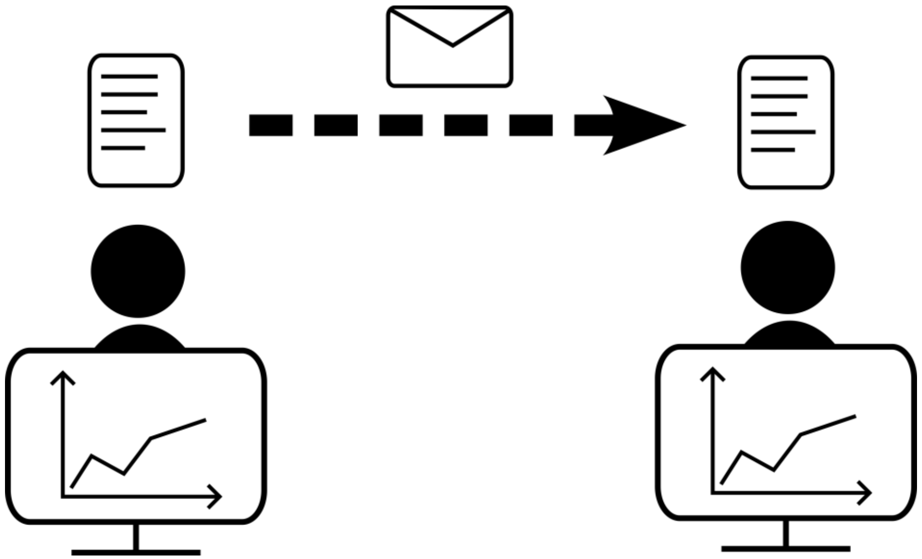
# Managing **INTEROPERABLE** Data

In what ways and why is data interoperability arguably the most complex aspect of FAIR practice?

- ✓ File Formatting + Types
- ✓ Metadata

# What is data interoperability?

The ability to exchange and use data across multiple systems



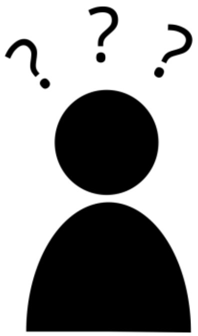
The ability of diverse datasets to be merged and aggregated in meaningful ways

- 0.5L milk
- 3 eggs
- 150g flour

+

- 1 gallon milk
- 200g egg
- 3oz flour

=



# File Types and Formatting

# FAIR file formats in biological practice

1

Use .csv or .xls files for numerical data. **Never** share data tables as word or pdf

2

Provide underlying numerical data for all plots and graphs

3

Use common file formats (domain specific)

4

Convert proprietary binary formats to open ones. For example convert Snapgene to Genbank/SBOL, microscopy multistack images to OME-TIFF

# Data formatting

- **Use standard formats wherever possible**
  - Dates in ISO8601 format: 2025-10-16 14:53:02
- **Be consistent!**
  - For a variable with two levels: TRUE/FALSE, 0/1, Yes/No,…?
  - Use the same units (often standard within a field)
- **Think ahead**
  - Every time you change your data formatting, you will either break interoperability with previous data, or generate work to re-format it

# Interoperable Metadata

# Metadata

**Think as a consumer** of your data, not the producer!

What information would another researcher need to understand or reproduce your data?

- Name of data creator/ Title
- Project purpose and experimental hypothesis
- Biological material, context, details on data acquisition, experimental conditions, etc.

+ Other datasets and publications associated with this image/ dataset (DOIs, links, etc).

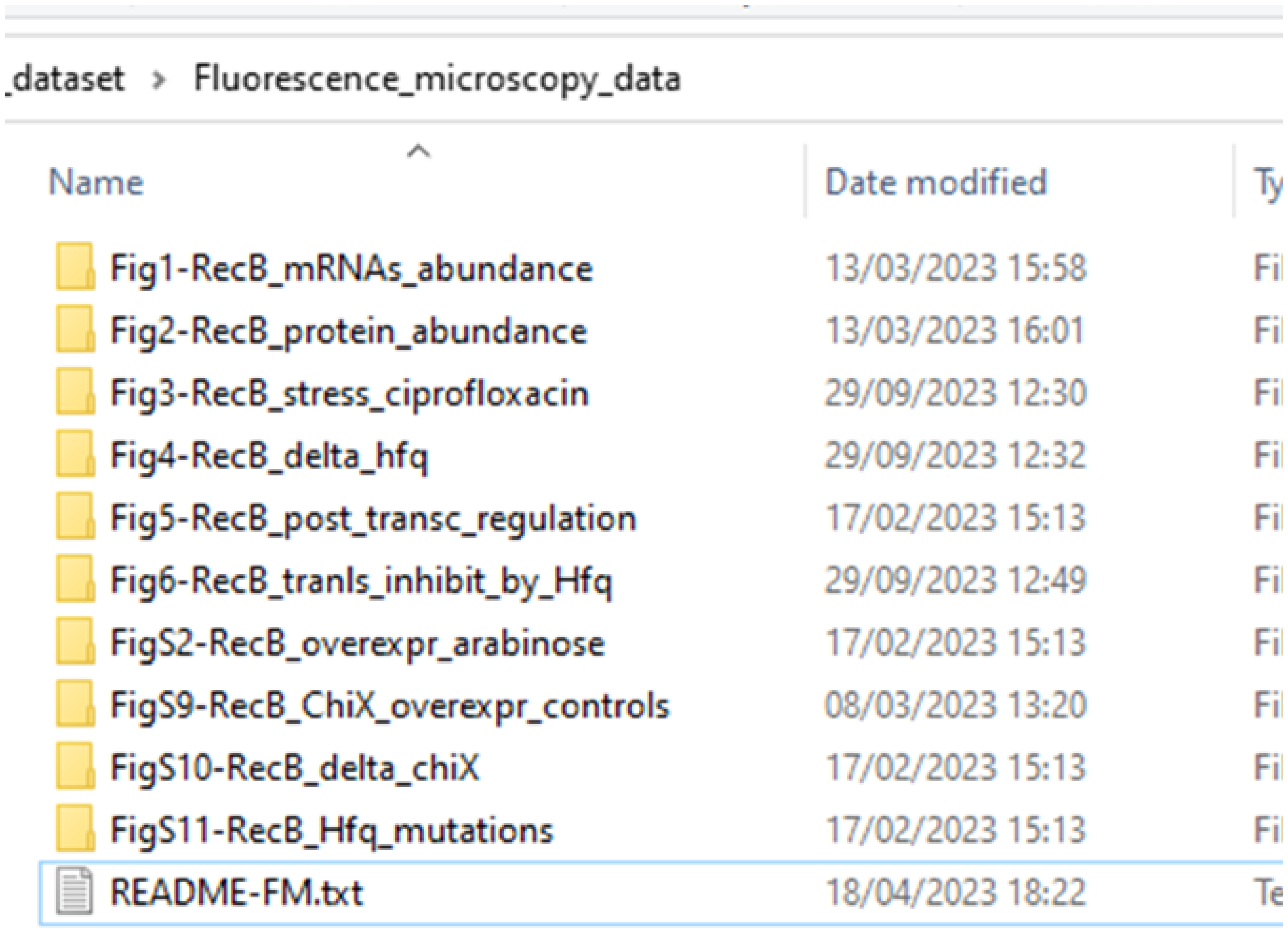
# Metadata

| | A | B | C | D | E | F |
|---|---|---|---|---|---|---|
| 1 | **Title** | Clock modulation of starch, pigments and nitrogen | Timet Project | | | |
| 2 | **Funder** | BBSRC-BB-V030557-1 | | | | |
| 3 | **Authors** | Sosa M | Pintos A | | | |
| 4 | **Contact** | Thee Bosss bio_rdm@ed.ac.uk | | | | |
| 5 | **Growth protocol** | https://doi.org/10.21769/BioProtoc.126 | | | | |
| 6 | **Measurements protocols** | https://doi.org/10.1038/nprot.2006.232 | https://doi.org/10.1038/nprot.2009.12 | | | |
| 7 | **Study date range** | 2019-12-05 to 2020-01-05 | | | | |
| 8 | **Cabinet** | Percival E-36L | | | | |
| 9 | **Data description** | **Metabolites reported per g of fresh weight of 6-week-old plant leaf rosettes** | | | | |
| 10 | **Data dictionary** | | | | | |
| 11 | **SD** | short days 6 h light | | | | |
| 12 | **LD** | long days 18 h light | | | | |
| 13 | **+SUC** | 10 mg sucrose added per GM-agar | | | | |
| 14 | **FW** | fresh weight | | | | |
| 15 | **Kegg IDs** | | | | | |
| 16 | **Starch** | C00369 | | | | |
| 17 | **Sucrose** | C00089 | | | | |
| 18 | **Cholorophyll** | C01793 | | | | |

metadata | data

Metadata in Excel

# Metadata





Metadata in
README files

# Metadata

This documentation file was generated on 2022-05-20 by Ines Boehm (https://orcid.org/0000-0002-9343-0944)

## Contextual Information

### Repository description

Here you will find code chunks used to analyse neuromuscular junctions (NMJs) for example their morphology, and resulting data.

### PURPOSE

This repository has been generated to share code, code snippets and data used in NMJ- analysis (e.g. the comparative anatomy of the neuromuscular junction between mammalian species).

### ETHICS

All tissue from human studies were covered by the requisite ethics approvals (NHS Lothian REC: 2002/1/22, 2002/R/OST/02; NHS Lothian BioResource: SR719, 15/ES/0094).

## Repository file overview

| Filename | Description |
|---|---|
| dendogram_distance_matrix.R | script to generate a Tanglegram |
| PCA_muscavr.csv | average data from NMJ-morph/aNMJ-morph for each muscle from all species analysed |

### KEYWORDS

neuromuscular junction, NMJ-morph, aNMJ-morph, morphology, extensor digitorum longus, peroneus longus, soleus

### DATA REUSE

This repository is of interest for (a) beginners wanting to familiarise themselves with NMJ-analysis. e.g. NMJ-morph (https://doi.org/10.1016/j.celrep.2017.11.008) or aNMJ-morph (https://doi.org/10.1098/rsos.200128) as the raw data provides the opportunity for comparison; (b) researchers in the field of comparative anatomy wanting to compare their species' NMJ-morphology to other species average NMJ morphology;

## General Information

### Biological Context

The neuromuscular junction (NMJ) is a synapse formed between lower motor neuron and target skeletal muscle fibre. Due to its crucial role in the facilitation of movement, it is involved in neurodegenerative conditions and both basic neuroscience and clinical neuroscience research focus a lot of their efforts on its study. Whilst the NMJ is affected in many neurodegenerative conditions in humans, the vast majority of

https://github.com/Boehmin/NMJ_analysis/blob/main/README.md

TXT

Metadata in README files

# Exercise #3

**Resource**: File examples

- bit.ly/4nVUKvz

**Task A**: Which of these file formats are interoperable or not?

- Try to copy the "Supplementary_table.pdf" data into Excel
- Try to open "SI_fig_1.opj"
- Is "Fitting_model.png" interoperable?
- Is the example Python script interoperable?
- Open the "Ens_data.xlsx" file: is it interoperable?

**Task B**: What alternative file formats would increase interoperability?

**Output**: Context about the interoperability of different file formats

# Enduring Understanding

At any scale, data interoperability can be logistically challenging (i.e. between different software) and conceptually challenging (i.e. removing data from its original context may yield different findings).

# Practical Applications

- Let other people reuse your data (and cite you for it!)
- Facilitate collaborations
- Keep access to your data if you lose access to software

# Managing **REUSEABLE** Data

To what extent is reusable data consequential and for whom?

- ✓ Data Management Plans
- ✓ Research Outputs
- ✓ Data Sharing Cycle

# Data Management Plans

Reusable


**Plan ahead**

Data management plans (DMPs)

*Figure credits: Tomasz Zieliński and Andrés Romanowski*

Ed-DaSH

# Data Management Planning: What? How? Where?

**What:** thinking about what data you will generate.

✓Types of data, format, and volume of the data

**How:** thinking about how you will:

✓Organise, describe, securely store, and back up your data.

**Where:** to ensure your data is available for replication and reuse.

✓Archiving, publishing, and conditions of re-use.

# Writing a DMP for your Research

## Tool: DMPOnline https://dmponline.ed.ac.uk

## Research Data Services (RDS) Hands-on workshop(s)

- Understand the necessity/benefits of producing a DMP.
- Discover how to register for and use DMPonline.
- Draft a basic DMP!

https://library.ed.ac.uk/research-support/research-data-service/research-data-training-skills/workshops-courses



# BioRDM DMP advice

https://www.wiki.ed.ac.uk/display/RDMS/Data+Management+Plans

# Research Outputs

# FAIR in biological practice

- Describe your data well (good metadata).
- Write a README file describing the data.
- Use descriptive column headers for the data tables.
- Tidy data tables, make them analysis-friendly.
- Use standard (meta)data formats (e.g. SBML, SBOL).
- Use commonly known terms and PIDs in descriptions.
- Attach license files (for research data, normally CC-BY).

# For your narrative CVs…

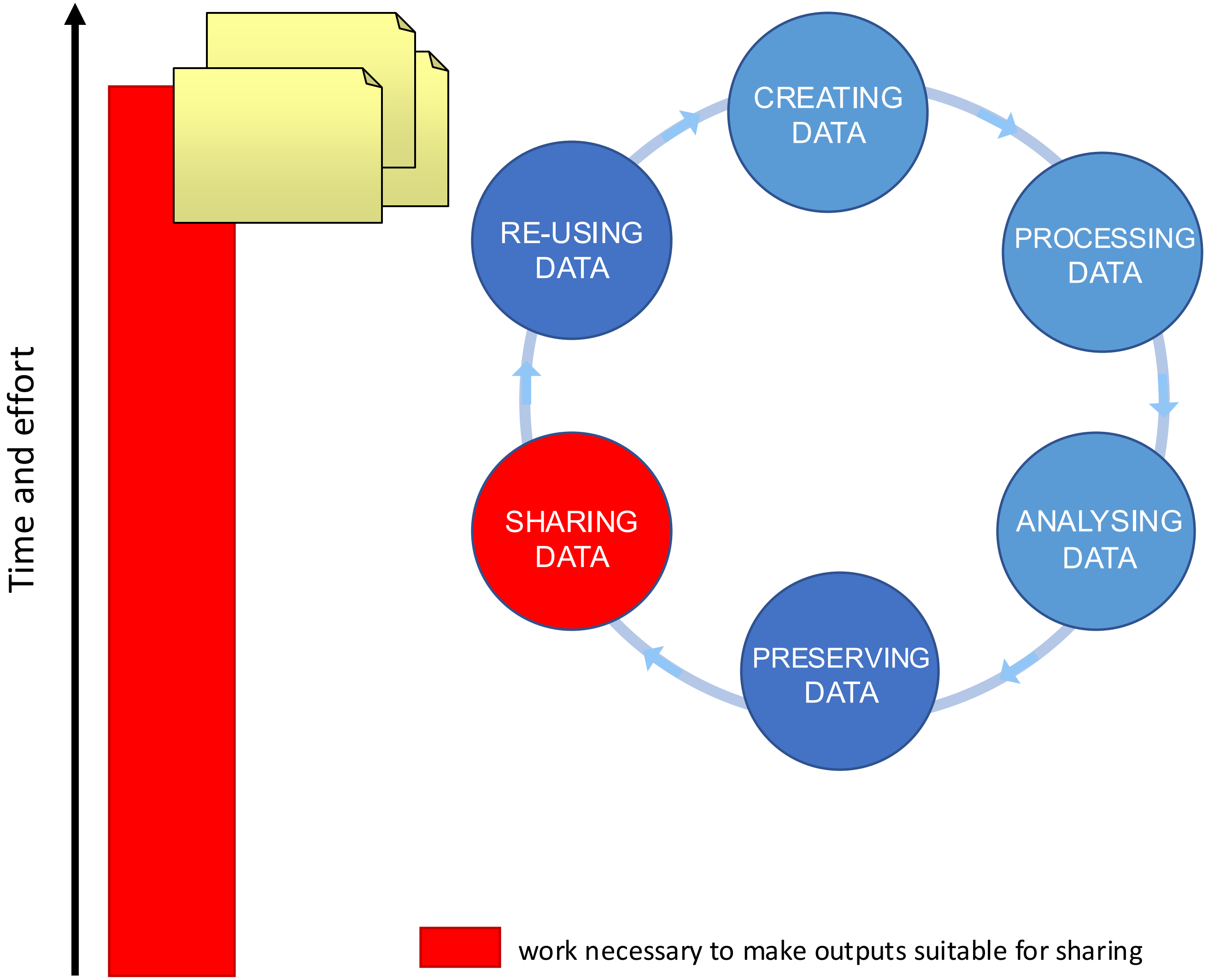
**… you need to have as many public outputs as possible**:

- Datasets in public repositories
- Public protocols
- Code packages
- Public engagement
- Involvement in community groups, student groups etc.
  (BioPod, Doors Open Day, Festivals, etc)

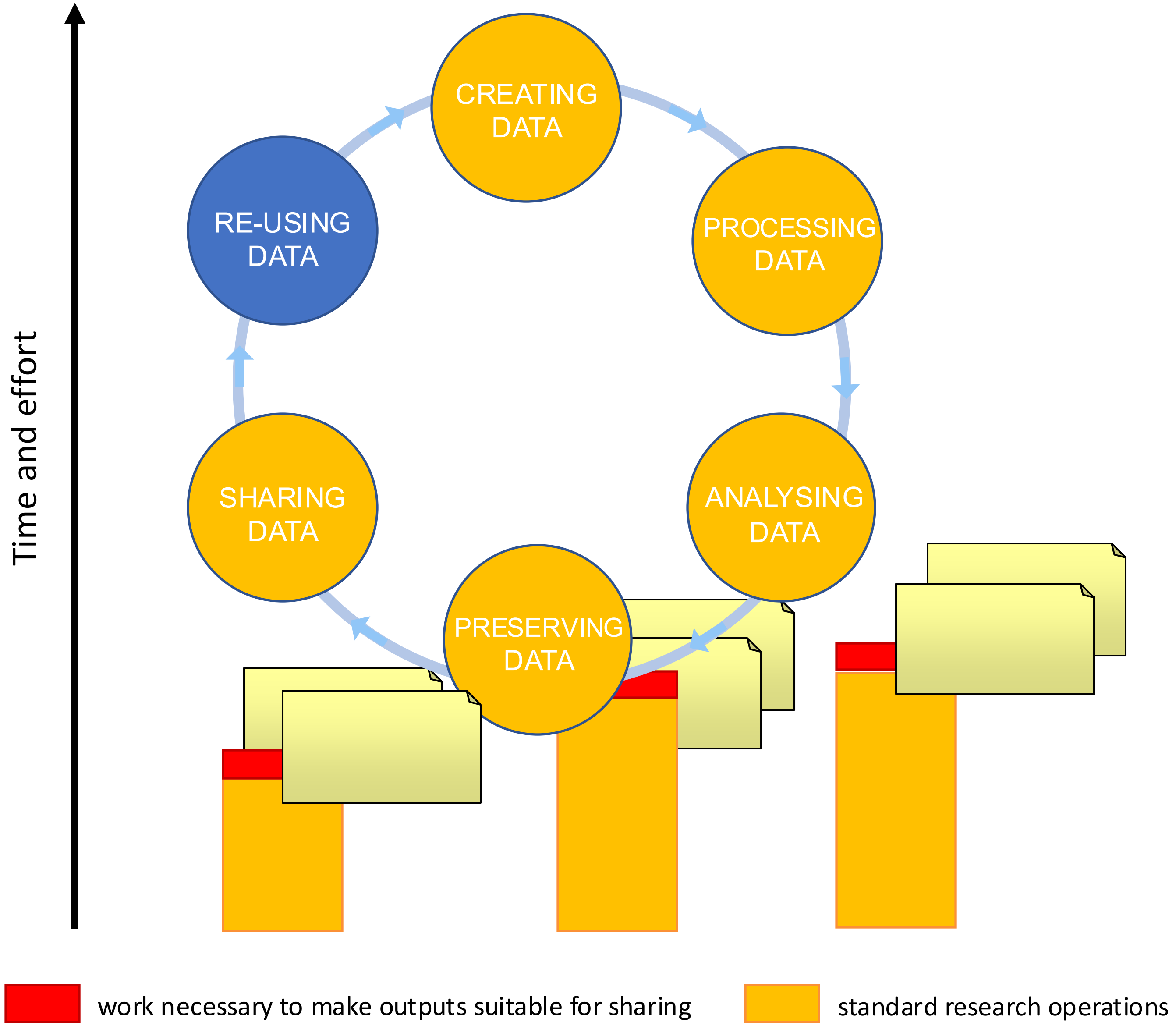
# Data Sharing Cycle

Reusable

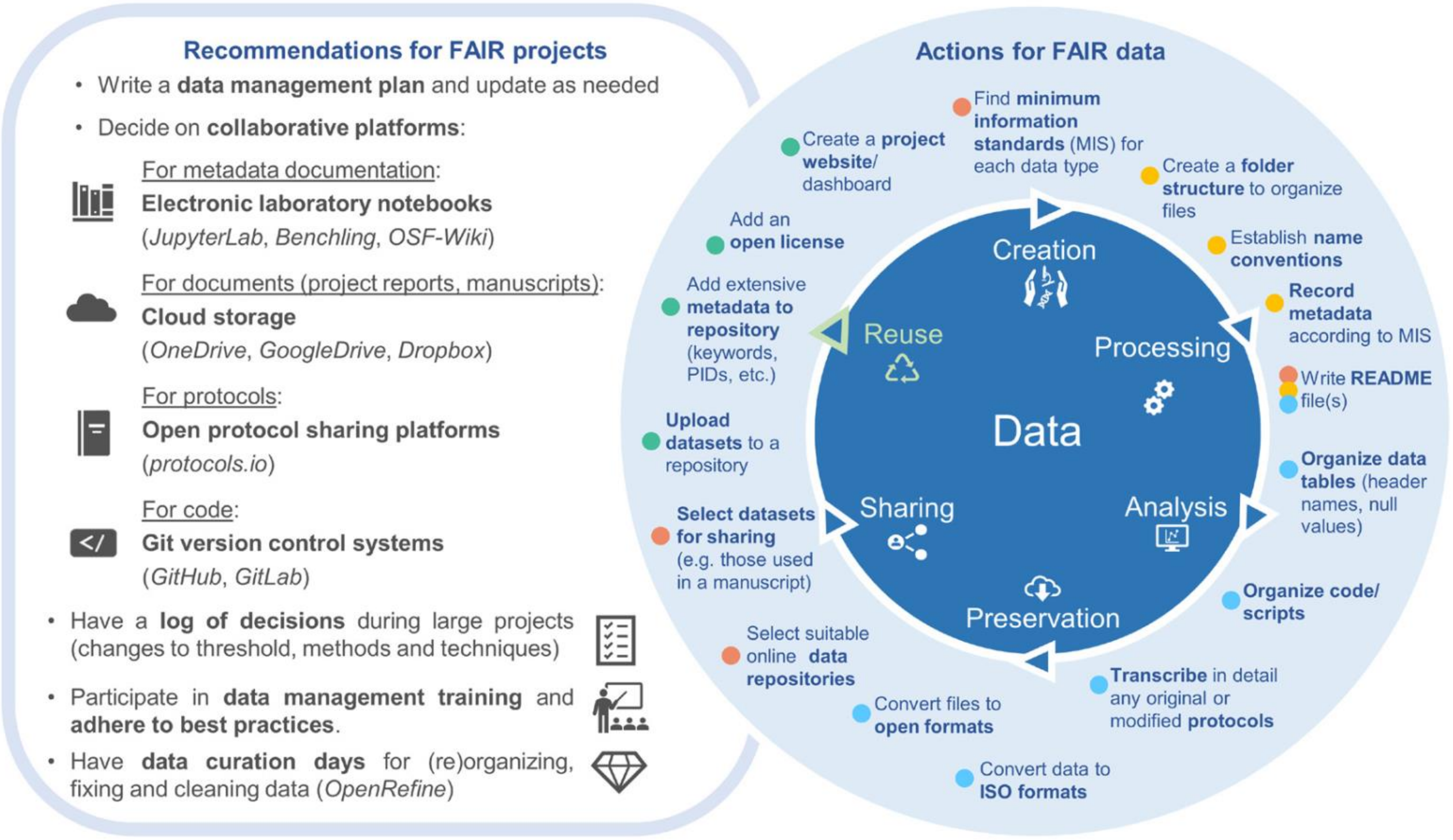
# Sharing as part of the workflow



# Sharing as part of the workflow



# Summary




Scorza LCT *et al.* Daily life in the Open Biologist's second job, as a Data Curator- Wellcome Open Res 2024, **9**:523 (https://doi.org/10.12688/wellcomeopenres.22899.1)

Reusable

# Exercise #4

**Resource**: PhD project background info

**Task A**: Answer or reflect on the following questions

- What data types do you expect to generate?
- What file formats will you work with?
- How much data do you expect to generate over your whole PhD?
- Where will you store the data?
- How will you transfer data (e.g. to collaborators)?
- What folder structure will you use for your projects?
- Will the data be made publicly available at the end of your project?
  - If yes: on which repository could you make the data available?

**Output**: Preliminary Data Management Plan

# Enduring Understanding

Reusable data is the cornerstone of data sharing, and promotes collaboration and career progression.

# Practical Applications

- Make it easy to share data at the end of a project
- Support your CV by sharing meaningful outputs
- Contribute to data re-use and experimental reproducibility!

THE UNIVERSITY *of* EDINBURGH
Digital Research Services

# Navigating the Digital Research Services' Support and Activities

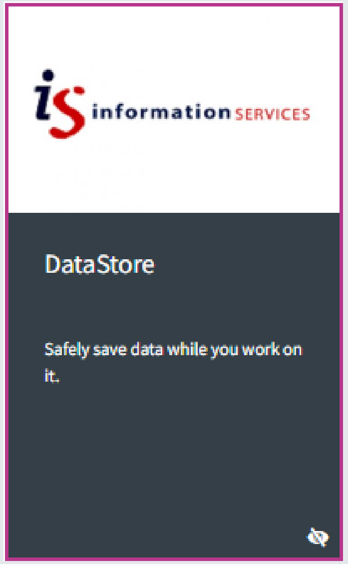
**School of Biological Sciences**
Dr André Vieira | Research Facilitator, CSE

**How can the DRS help?**

We act as a focal point for all the data and computing services available to researchers at the UoE.

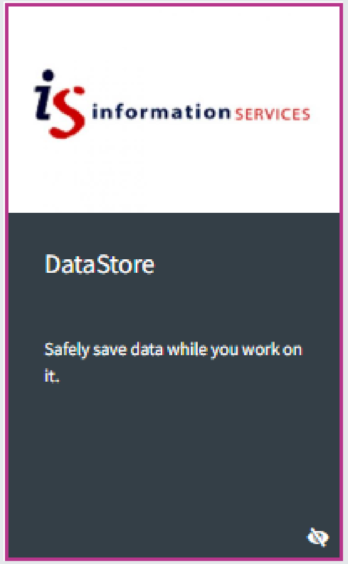

JOURNAL CHECKER TOOL
Journal Checker Tool
Which publishing options are supported by your funder's OA policy?
Overleaf
Overleaf
An online LaTeX writing tool that allows you to easily create and collaborate on formatted scientific and technical
information SERVICES
Eddie
A high-performance Linux computing cluster to run analysis faster than a desktop computer.
Edinburgh DataShare
DataShare
Share your data with other researchers. Make your data FAIR!
Data Asset Framework
EIDF Edinburgh International Data Facility
CLOCKSS
CLOCKSS
An archive to ensures the long term preservation of and access to digital content.
Edinburgh Research Office
Get support in identifying funding, crafting your application and managing
Digimap
EDINA
Geospatial Consultancy
Consultancy for geospatial technology applications, use and advice e.g. GIS, mapping
UK Data Service Census Support
A central clearing house for UK pop census data from 1971 to present.
An online map and available by subscri establishments.
Open Science Framework
An open source project management platform for researchers.
Publish and Share
Office Equipment
General Computing
Telephony, Printing and Scanning
CMVM Research Computing Overview
Edinburgh Diamond
A service provided by Edinburgh University Library that supports the publication of academic and student-led
archer2
ARCHER2
ARCHER2 is the UK National Supercomputing Service.
era
Edinburgh Research Archive (ERA)
Need a Digital Object Identifier (DOI) for your project report or briefing paper? Deposit your research output in ERA.
Pure
Create a record of all your research activities, data and outputs to showcase your research work.
Statistical Accounts of Scotland
Digitised and searchable versions of the Old Statistical Account and the NEW Statistical Account.
DataVault
Store your research data for the long term, safely and privately, in DataVault.
GitLab
TDM ON

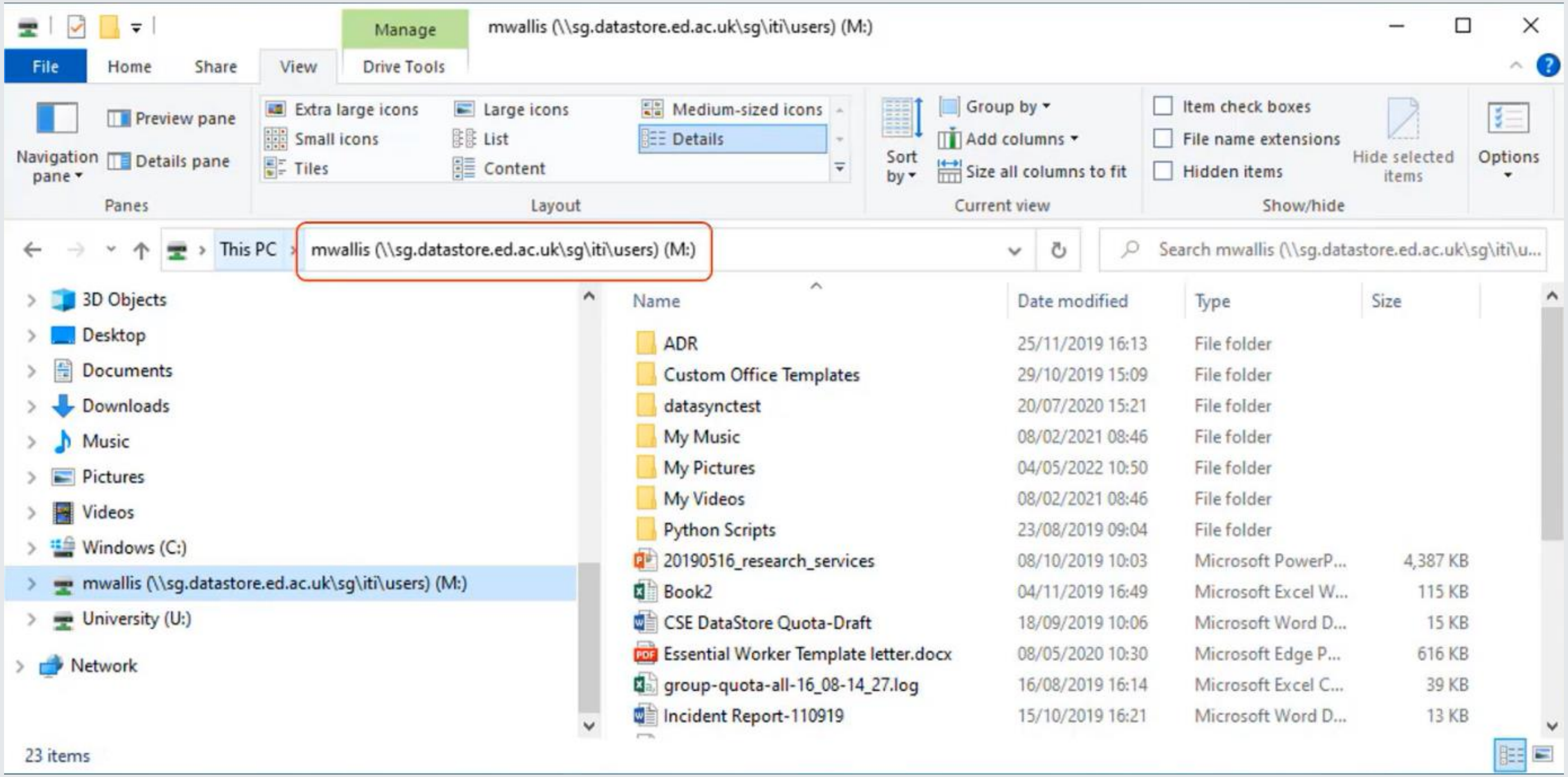
# SOME DIGITAL RESOURCES



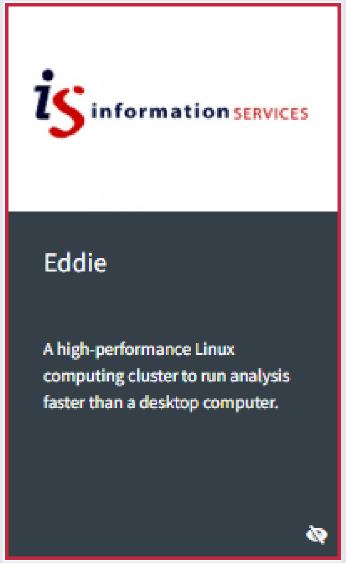

- Free 'at point of use' individual allocation of 500 GB for every researcher
- Group spaces (One or more people)
- Backed up to 3 different locations with offsite disaster recovery platform and offsite backup and retention
- Restricted to University of Edinburgh computers
- Support for very large data (>1 PB)
- Connection between DataStore personal allocation or group space with DataSync
- No Data encryption

# SOME DIGITAL RESOURCES



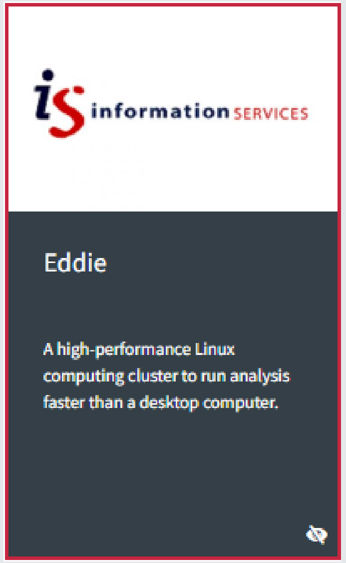



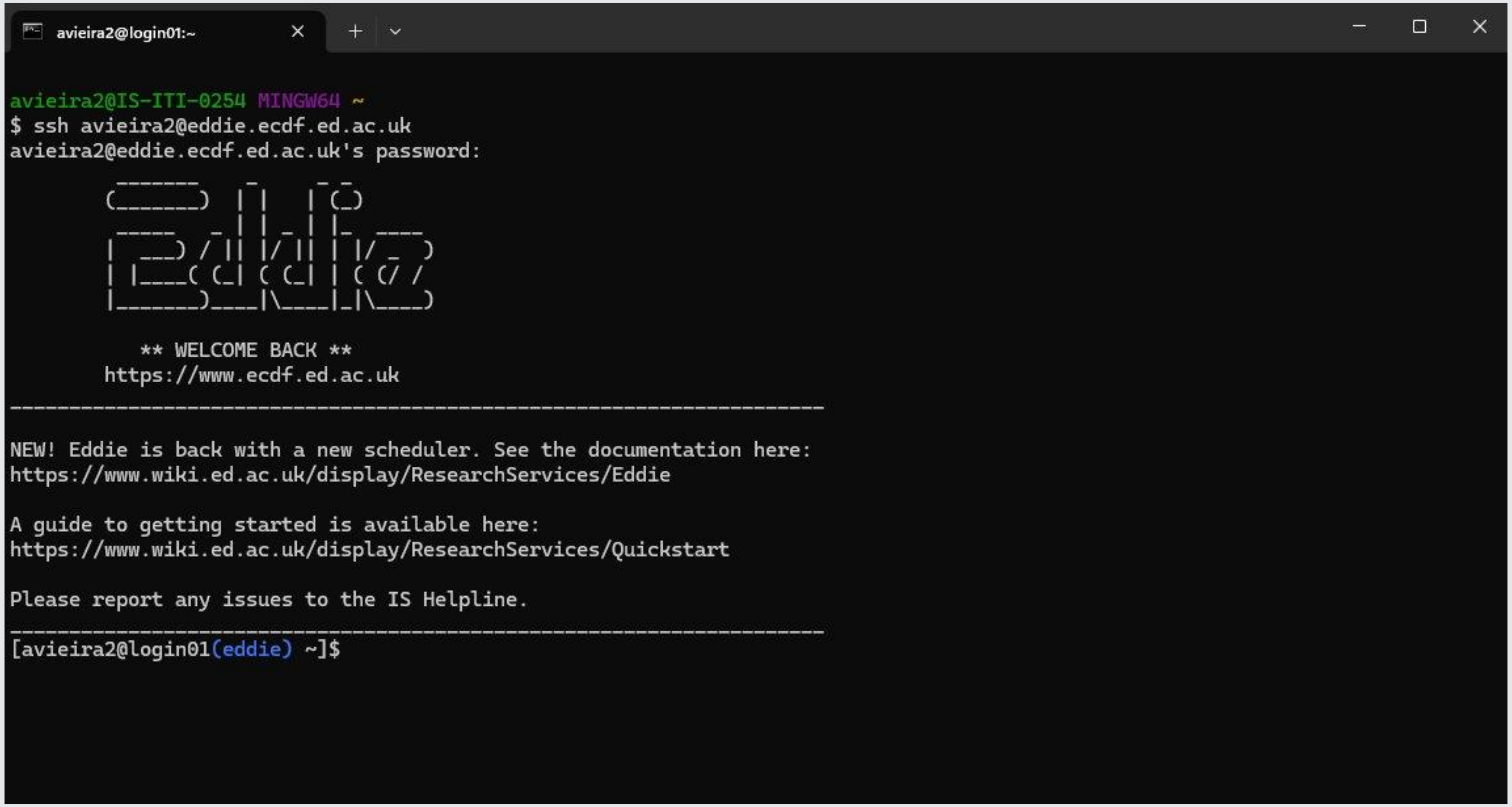
# SOME DIGITAL RESOURCES



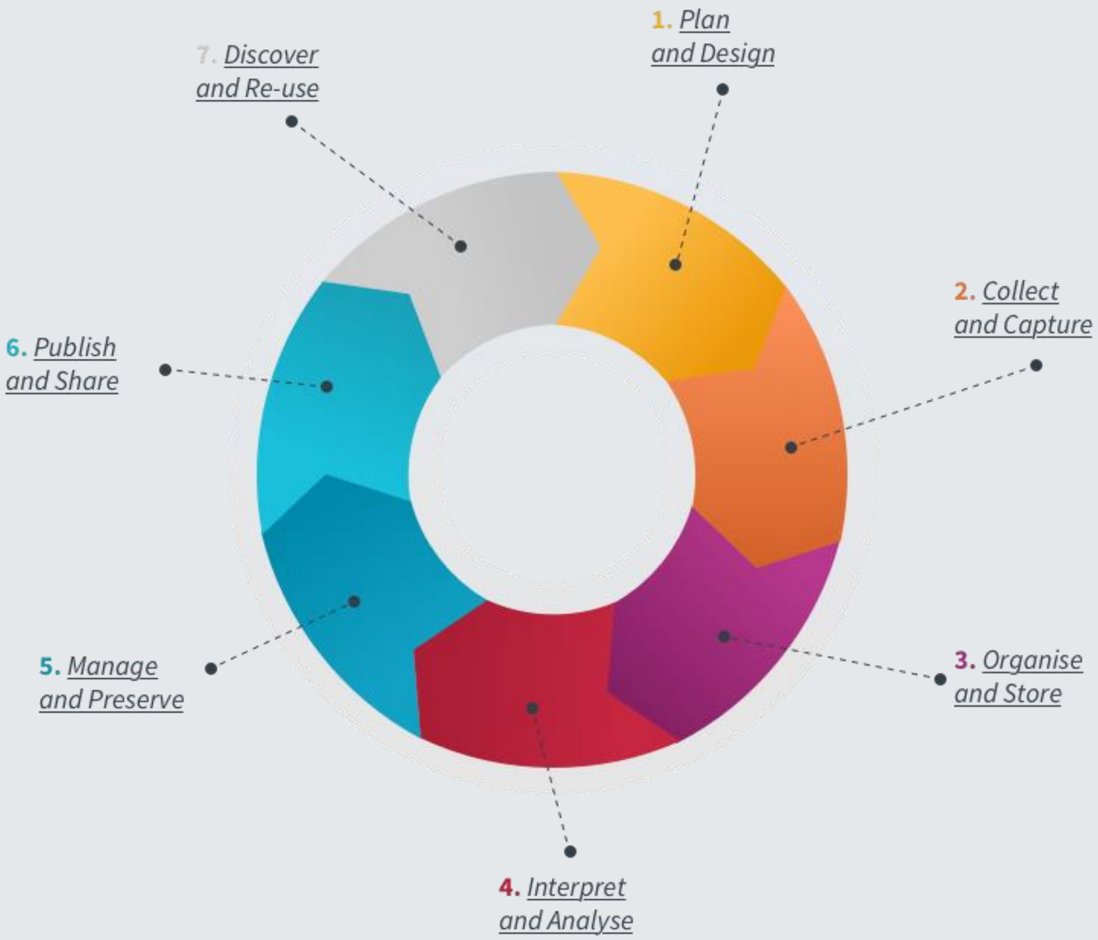

- **Work can be run in parallel taking advantage of multiple compute cores**
- **Priority and exclusive access to resources is available for funded projects**
- **A number of applications are installed on Eddie and you can install your own**
- **Eddie connects with DataStore**
- **Training, support and consultancy services are available**
- **Standard allocation: Free**
- **Paid options:**
  - Priority
  - Ringfence - Exclusive access to one or more of the computing nodes
  - Storage

# SOME DIGITAL RESOURCES



# DIGITAL RESEARCH SERVICES




## THE DRS CAN HELP RESEARCHERS WITH:

- RESEARCH PLANNING
- DATA STORAGE AND DATA SHARING
- COMPLY WITH BEST RESEARCH PRACTICES
- SPEED UP ANALYSIS
- MEETING FUNDER REQUIREMENTS
- PUBLISHING AND PROMOTING YOUR DATA AND WORK – OPEN RESEARCH

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Digital Research Services

Resources | Resource Finder | Research in Action | Develop Your Skills | News and Updates | Who Can I Contact | About us

# Digital Research Services

A single point of access to all of the data and computing services available to researchers at the University of Edinburgh.

FEEDBACK

## Working with you throughout the research life cycle

Use the interactive lifecycle to access resources relevant to each research stage.


### I am looking to perform the following task(s)

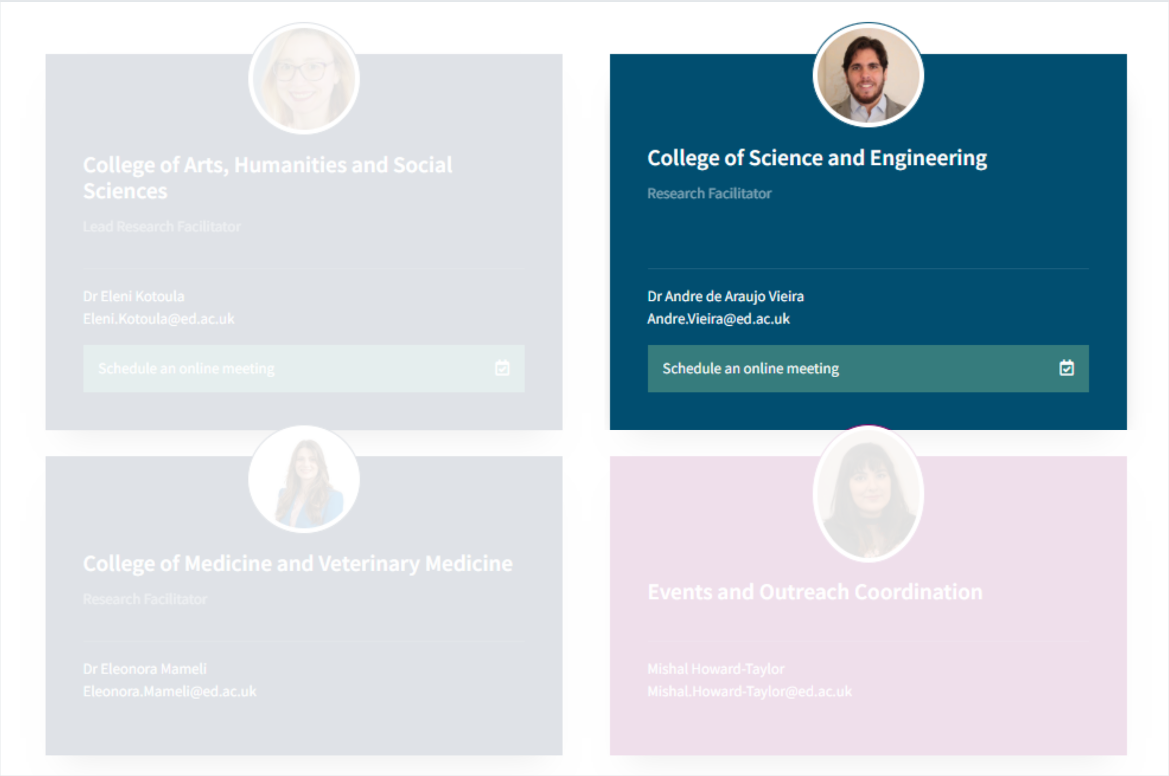
- Finding data
- Gathering data

# RESEARCH FACILITATION TEAM

Connect researchers and service providers

Community Building & Networking

Enhance the research with digital resources




Help you navigate the complex landscape of digital tools

Tailored approach to the problem

Produce digital resource-related training material

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# Cultivating Digital Growth

Amplify your research efforts by participating in our fully funded internship scheme for postgraduate research students, developed by Digital Research Services.

**More info:** edin.ac/DRSAmbasssadors

# 2025 DRS AMBASSADORS PROJECTS, HOSTS & INTERNS

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| Projects | Ambassadors | | |
|---|---|---|---|
| | Ambassadors | College | Affiliation |
| Dashboard for research in Spatial Epidemiology, proposed by Glenna Nightingale (School of Health in Social Science) | Ezekiel Okeleye | CMVM | Veterinary |
| Development of a Graphical User Interface (GUI) for Training Deep Learning Models for High-Throughput Single Cell Imaging Data Analysis, proposed by Francois El-Daher (School of Biological Sciences) | Maryamu Habila | CMVM | Medicine |
| Development of R-Based Tools for Teaching and Training in the Usher Institute, proposed by John Wilson & Dr Keith Douglas (Usher Institute ) | Nicola Burns | CAHSS | PPLS |
| Development of Visualization Methods for Spikeinterface , proposed by Chris Halcrow (Centre for Discovery Brain Sciences ) | Huizi Zhang | CSE | Mathematics |
| Enhancing Secondary Mathematics through Climate Change and AI-Driven Games, proposed by Hui-Chuan Li (Moray House School of Education and Sport) | Zhang Zhaoxi | CSE | Mathematics |
| Fire Engineering & Physics-informed ML , proposed by Zak Campbell-Lochrie (School of Engineering) | Nishant Gaur | CSE | Engineering |
| Gender and Occupation Biases , proposed by April Bailey (School of Philosophy, Psychology and Language Sciences) | Ariadna Cervera | CSE | Informatics |
| Herodotus: ELM Automated Transcription, proposed by Huw Halstead & Dr Chiara Bonacchi (School of History, Classics and Archaeology) | Adam Koval | CSE | Physics and Astronomy |
| Improving Digital Research in Chemistry: Deployment of DataLab ELN , proposed by James Cumby (School of Chemistry) | Xi (Ian) Yang | CSE | Biological Sciences |
| LGBTQ Data Analysis , proposed by Kevin Guyan (Business School) | Milo Bischof | CSE | Geosciences |
| Research Facilitation , proposed by Eleni Kotoula, Dr Eleonora Mameli & Dr Andre Vieira (Research Services) | Balazs Szendrei | CAHSS | LLC |

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**DRS Podcast:**

From chemistry labs to classrooms, hear how DRS Ambassadors are driving innovation with digital tools, researcher engagement, and AI-driven learning.

# Ian, Balazs & Zhaoxi


RØDE

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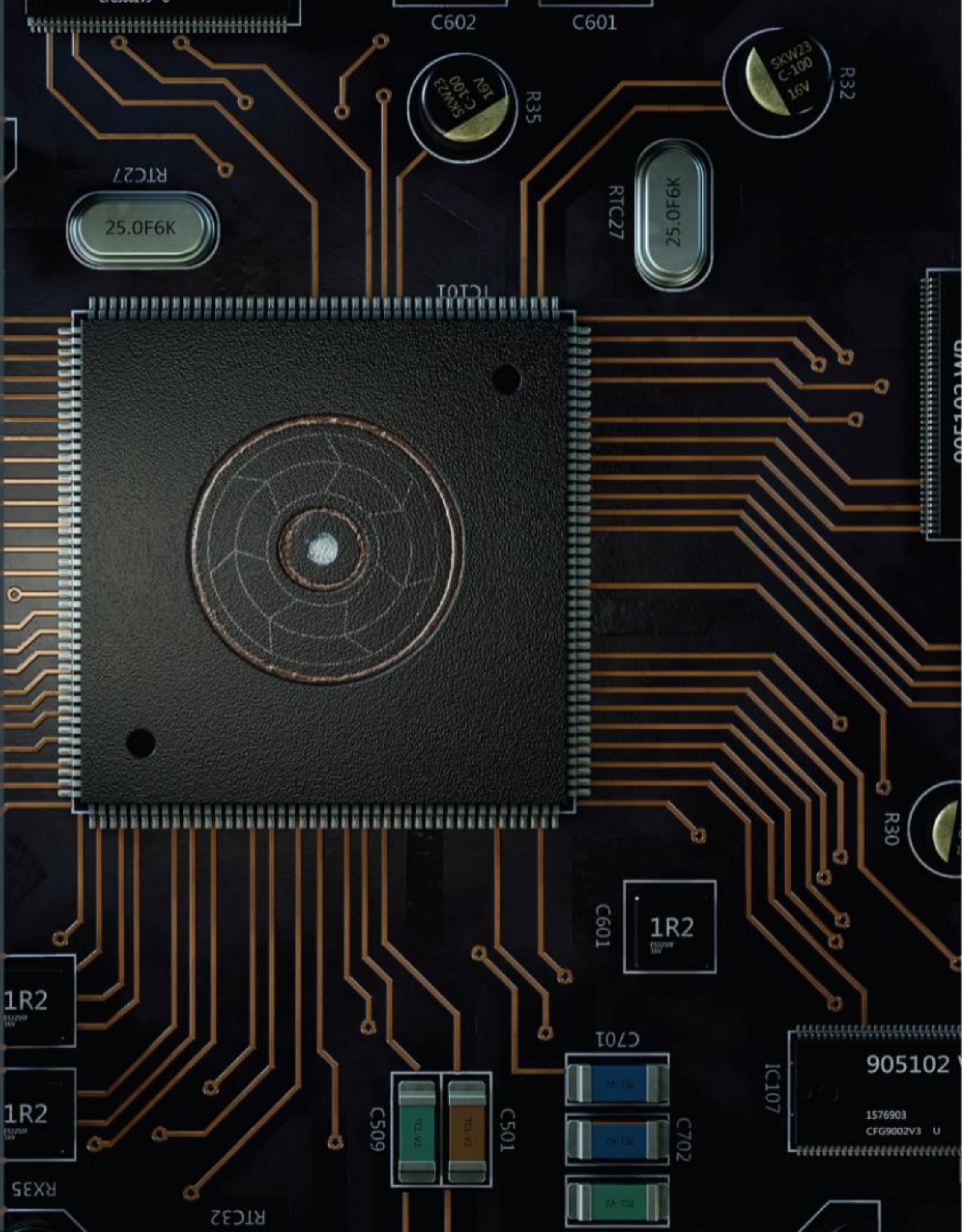
# HPC IN FOCUS

**Explore High-Performance Computing in a series of events Online and at the University of Edinburgh King's Buildings Campus.**

https://digitalresearchservices.ed.ac.uk/training/hpc-in-focus



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# HPC IN FOCUS

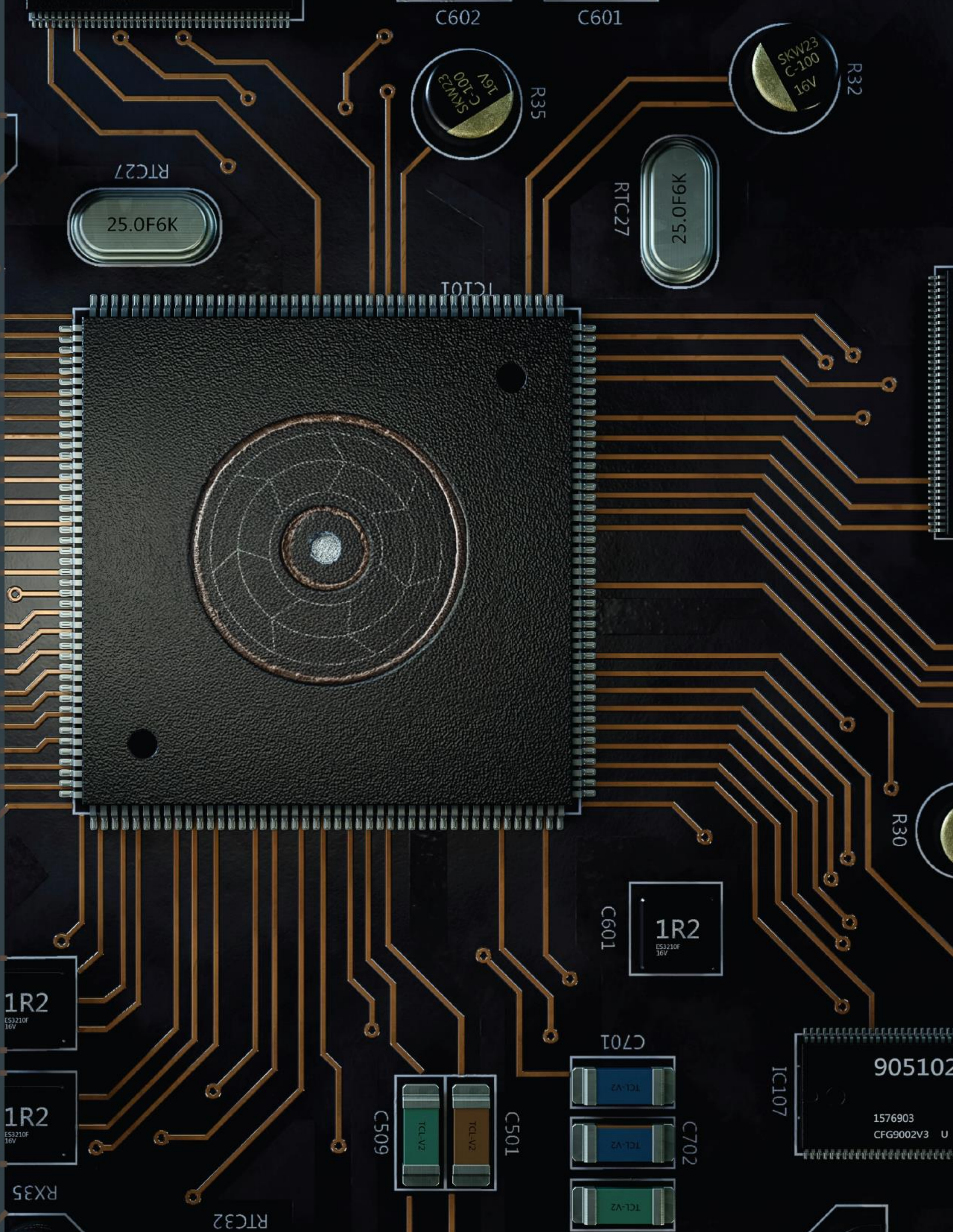
**02/10 – *Online, MS Teams, 15:00 to 16:00***
INFORMATION SESSION
What HPC Services does the University offer?

**09/10 – *In-Person, Swann Building, Room 7.15,15:00-16:30***
INSIGHTS, COMMUNITIES & NETWORKING
Relevant Communities to get involved

**23/10 – *In-person, Charteris Land, Room 4.04, 10:00 - 16:00***
HPC HANDS-ON TRAINING SESSION
Introduction to ARCHER2 – Part 1

**30/10 - *In-person, Charteris Land, Room 4.04, 10:00 - 16:00***
HANDS-ON TRAINING SESSION
Introduction to ARCHER2 – Part 2

**06/11 - *In-person, EFI, Room 1.52, 10:00 to 16:00***
HANDS-ON TRAINING SESSION
How to work on Eddie



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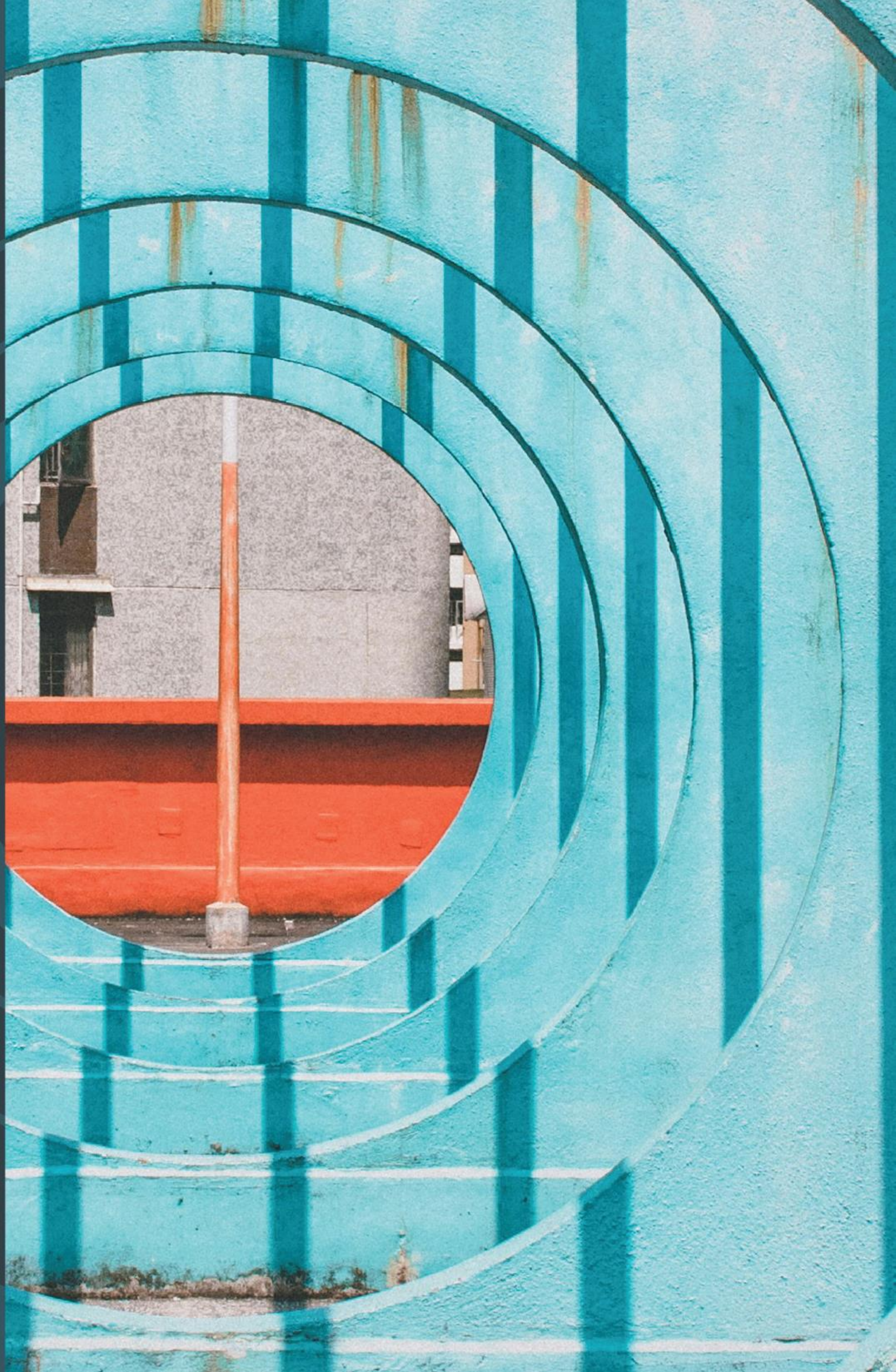
# Spotlight On:

## Research Planning

**Join our online seminar series, where we will tackle planning and designing your research.**



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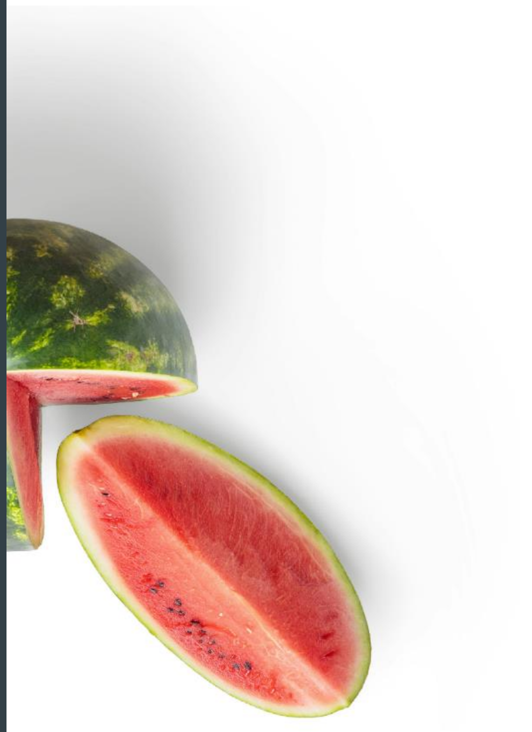
## DRS LUNCHTIME SEMINARS

# Enjoy a Slice of Digital Research

Discover key digital tools in your lunch hour.



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# CALL FOR ABSTRACTS NOW OPEN!

# Digital Research Conference

## 26TH FEBRUARY 2026

### THEMES

- AI in Research
- Digital Research Infrastructure and the Future of Research
- Computing Interdisciplinary Digital Research
- Ethics, Security & Integrity in Digital Research
- Green Digital Research Practices & Sustainability
- Embedding Digital Tools in Research, Innovation, Teaching and Learning

Find Out More:
**edin.ac/DRC**

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# KEEP IN TOUCH

https://mlist.is.ed.ac.uk/lists/subscribe/digitalresearch

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# Navigating the Digital Research Services' Support and Activities

School of Biological Sciences

Dr André Vieira | Research Facilitator, CSE

# In Conclusion

# Open Science

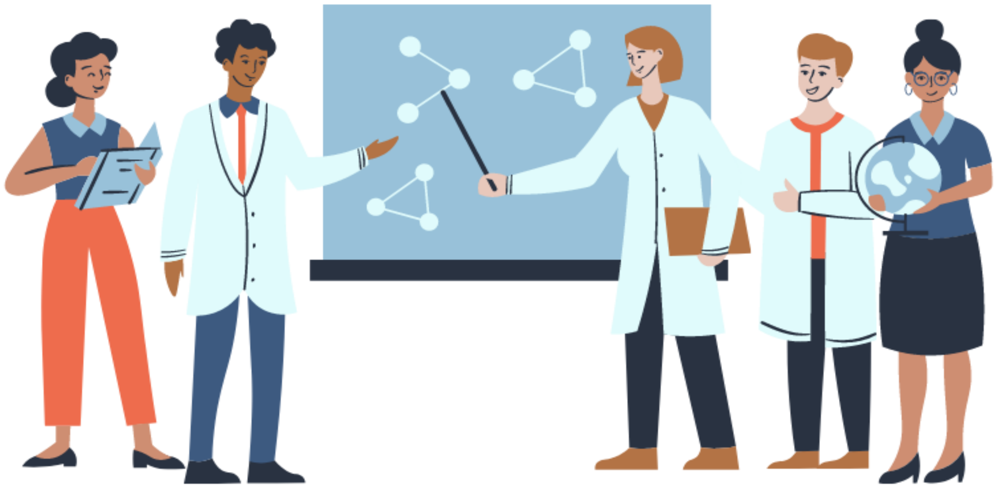
Open science is transparent and accessible knowledge

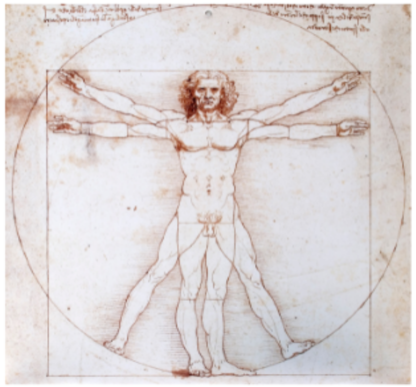
that is shared and developed

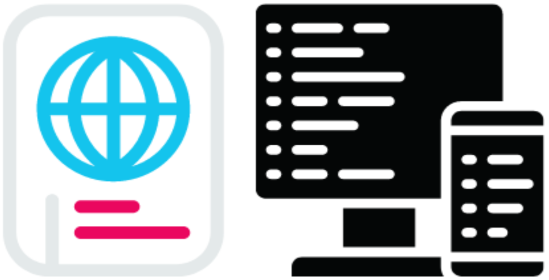
through collaborative networks.

Ed-DaSH

# Evolution of Open Science





Level of openness in research

# Open Science

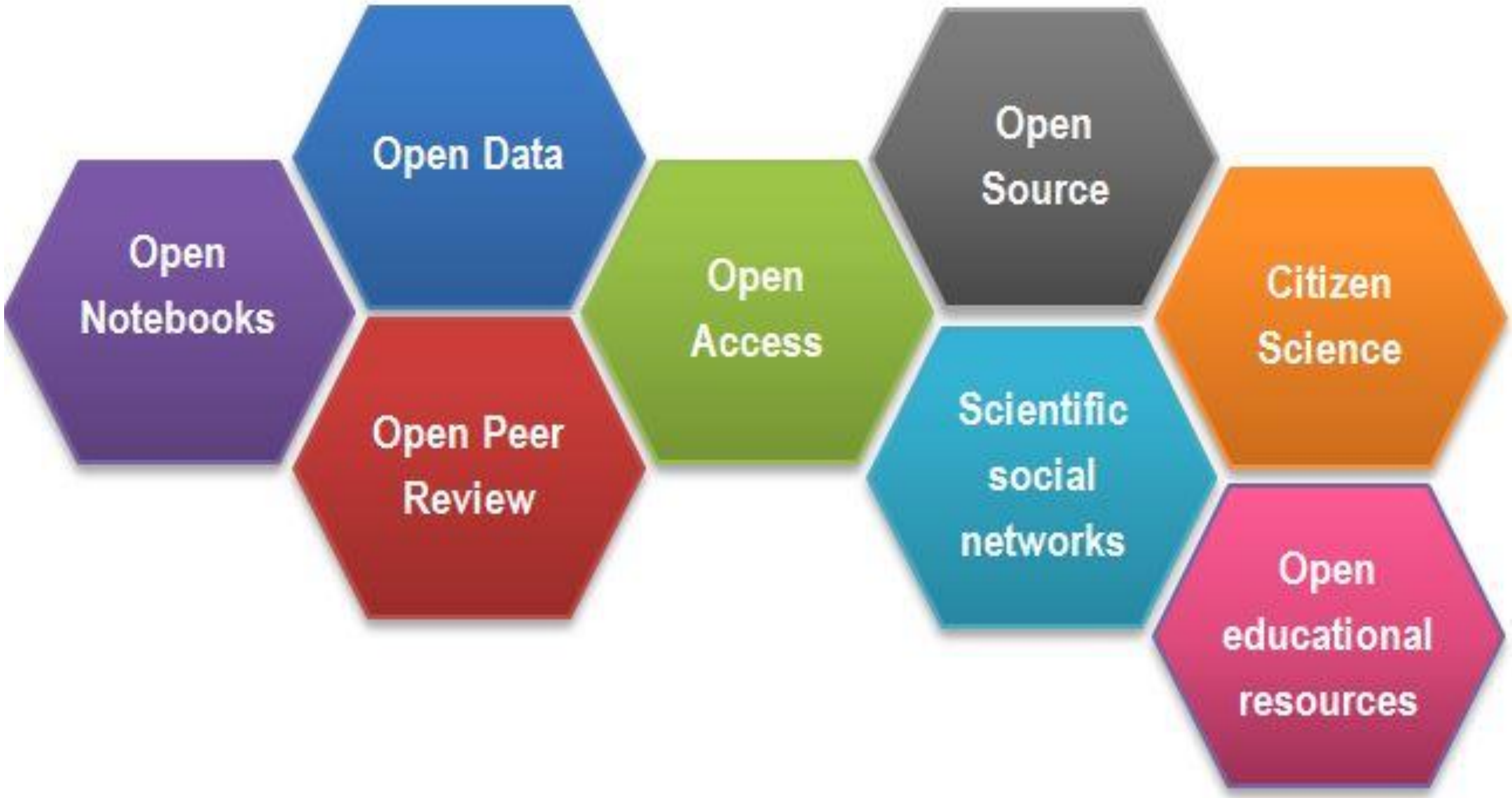
Open science is the movement to make scientific research **accessible to all levels** of society

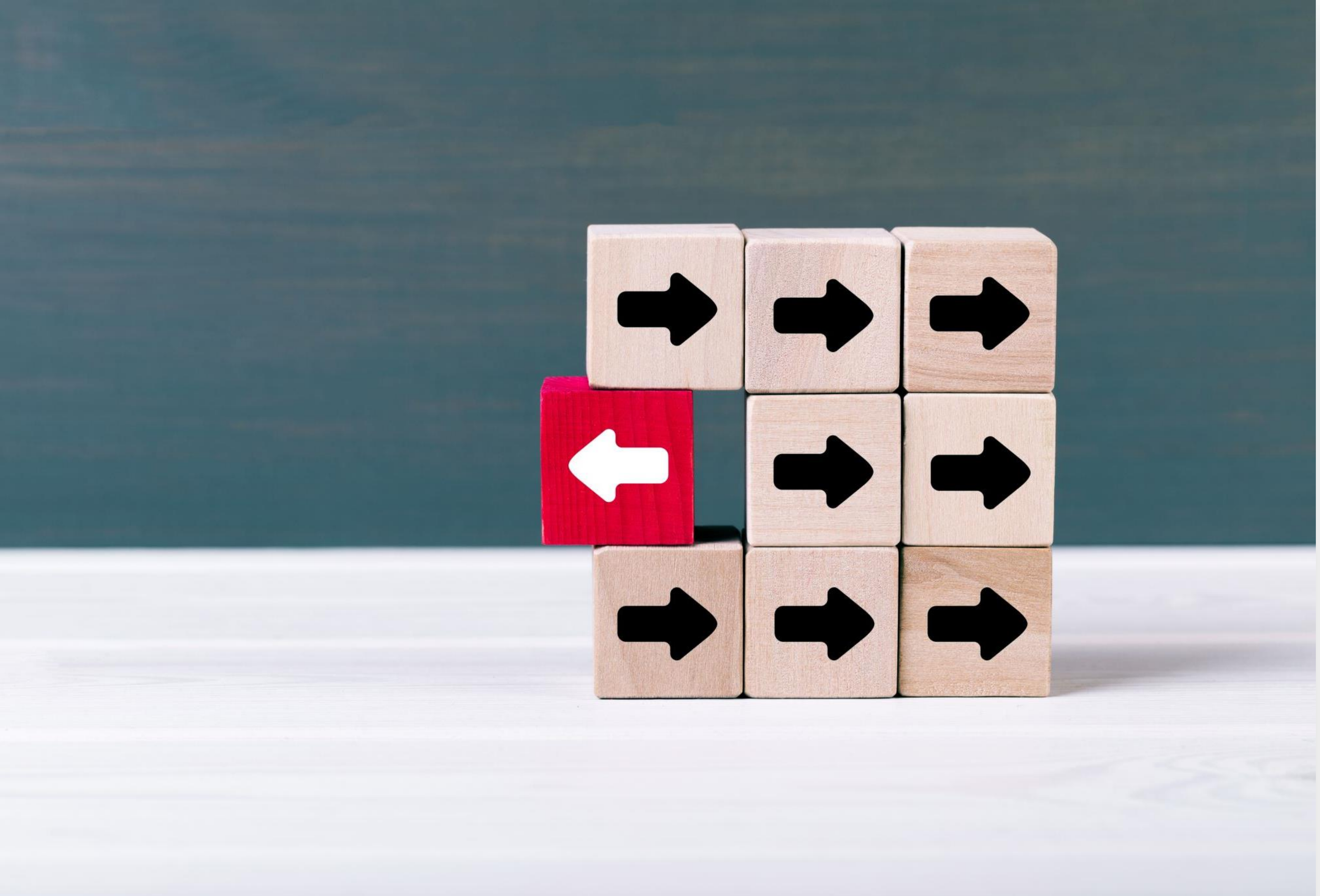
- publications
- data
- physical samples
- software

- general public
- amateurs
- professionals
- students
- teachers

**Digital technologies** for diffusing knowledge

# Open Science





# Outcomes of Openness

- Collaboration
- Speed
- Unpublished / negative results
- Re-use
- Learn by example
- Feedback

1 Go to wooclap.com

2 Enter the event code in the top banner

Event code

**IRJQNU**

Enable answers by SMS

Copy participation link

# Thank you!
# Best of luck in your PhD journey

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**Daniel Thédié, PhD. -** daniel.thedie@ed.ac.uk
**Juliana Rodríguez Cubillos. -** Juliana.Rodriguez@ed.ac.uk

**bio_rdm@ed.ac.uk**